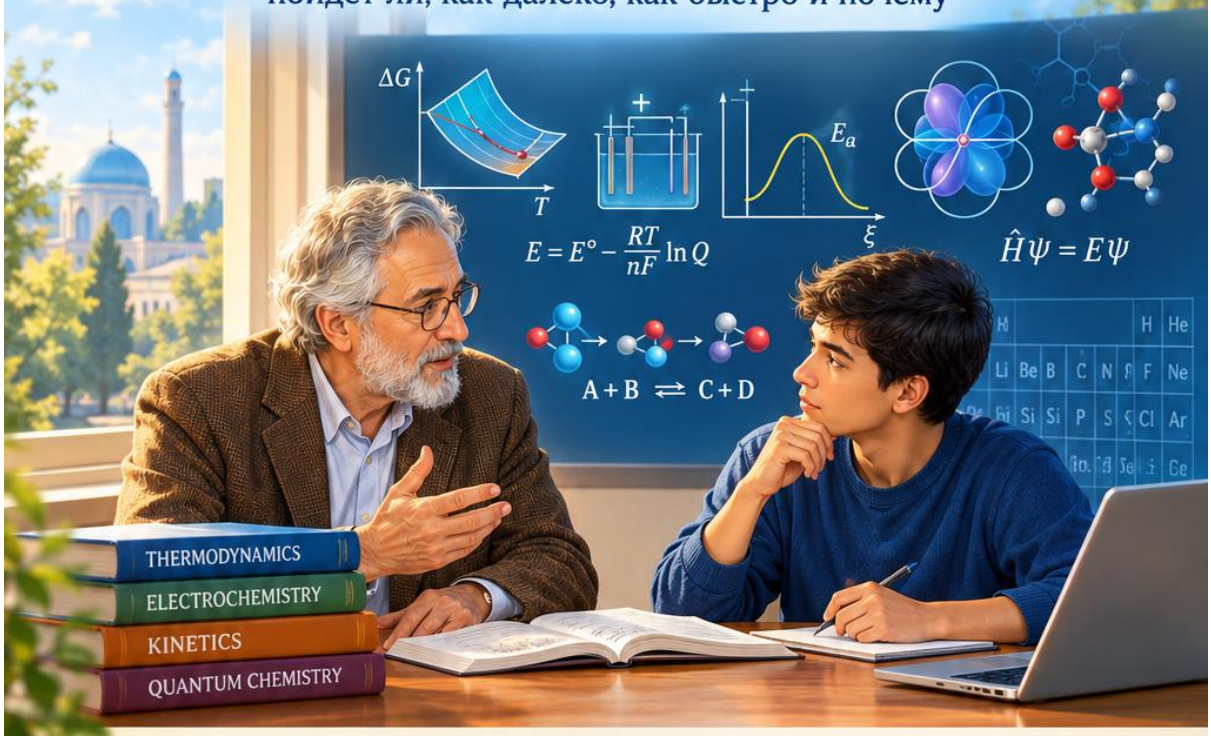
БЕСЕДЫ МУДРЕЦА И ЮНОШИ

ФИЗИЧЕСКАЯ ХИМИЯ

Рахимов Р.Х.

Термодинамика • Электрохимия • Кинетика • Квантовая химия

Как предсказывать химические процессы:
пойдёт ли, как далеко, как быстро и почему



PHYSICAL CHEMISTRY

Thermodynamics • Electrochemistry • Kinetics • Quantum chemiistry

How to predict chemical processes:
will it go, how far, how fast and why

12 глав • 31 рисунок • 25 таблиц
на русском и английском языках

12 chapters • 31 figures • 25 tables
in Russian and English

Серия посвящена памяти академика Садыка Азимовича Азимова

Tashkent • 2026

CONVERSATIONS OF THE SAGE AND THE YOUNG MAN

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The series is dedicated to the memory of Academician Sadik Azimovich Azimov.

Tashkent · 2026

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Preface

General chemistry answers the question **"what** will be produced?". Physical chemistry poses four other questions and answers them with numbers: **will** the reaction proceed (thermodynamics), **how far** will it go (equilibrium and electrochemistry), how **fast** (kinetics), and **why exactly this** way (quantum chemistry). Its language is energy, entropy, constants, potentials, and rates; its goal is to learn to predict, not just to memorize.

This book continues "Conversations of the Sage and the Youth: General and Inorganic Chemistry." We will not explain again what an atom, a bond, or pH is: all of this is already covered. We will elevate those ideas to a higher level. ΔG, K, E°, and activation energy, which were encountered there as "working formulas," will receive derivation, meaning, and limits of applicability here. The mathematics is at the level of algebra and logarithms; derivatives appear only as "rate of change," and each formula is accompanied by an example with numbers.

The form remains the same — a conversation. The sage knows where the formula comes from and where it **stops** working; the youth calculates everything himself, compares the result with the reference book, and notices when it is "too beautiful." If the youth's calculation diverges from the table — we do not hide the discrepancy, but investigate its cause: this is physical chemistry.

Structure of the book

Table 1. Plan of the book.

Part	Chapters	Contents
I. Chemical thermodynamics	1–4	First law and heat · entropy and second law · free energy, chemical potential, equilibrium · phases and solutions
II. Electrochemistry	5–6	Ions in solution · electrode processes, fuel cells and batteries
III. Chemical kinetics	7–9	Rate & order · temperature, Arrhenius, transition state · mechanisms & catalysis
IV. Quantum chemistry	10–12	Molecular orbitals & Hartree–Fock · potential energy surfaces & reaction calculations · spectra & photochemistry
Final conclusion	—	How the four parts form one picture

How to read: five markers

As in the entire series, each important statement has an honest epistemic status indicated by a colored frame.

[UV] Established fact
Repeatedly verified. Example: energy is conserved (the first law of thermodynamics).

[H] Hypothesis
Plausible, but not proven. Example: explanation of the low initial entropy of the Universe (chapter 2).

[NI] Not independently confirmed
It relies on one source or oral account.

[OD] Justified addition
Simplification or clarification from us: for example, the ideal gas model or a "reversible" process as an idealization.

[ERROR] Error or exaggeration
A common misconception that we correct. Example: "heat is what is contained in a body."

If the author's hypotheses (in particular, about the impulse tunneling effect) prove to be relevant in the physicochemical context, they will appear only with the label [Г].

Units and constants

Table 2. Constants we will use.

Quantity	Symbol	Value
gas constant	R	8.314 J/(mol·K) / J/(mol·K)
Boltzmann constant	kB	1.381·10−23 J/K
Avogadro's number / Avogadro constant	NA	6.022·10^23 mol^−1 / mol^−1
Planck constant	h	6.626·10−34 J·s / J·s
Faraday constant	F	96 485 C/mol
RT при 298 K	RT	2.479 kJ/mol
(RT/F)·ln 10 at 298 K	2,303RT/F	0.0592 V / V
1 eV per particle	1 eV	96.485 kJ/mol
1 calorie / 1 кал (термохим. / thermochemical)	cal	4,184 J / J

Physical chemistry is the chemistry that has numbers. Let this book teach the reader to trust the numbers — and to verify them.

PART I. Chemical Thermodynamics

Part I. Chemical Thermodynamics

Chapters 1–4

Chapter 1. Energy, Heat and Work: the First Law

In this chapter

System and environment, state functions · heat and work · first law · expansion work: reversible and irreversible paths · enthalpy and heat capacity · Kirchhoff's law · bomb calorimeter · why the heat capacity of hydrogen at 300 K is 5/2 R.

Система и окружающая среда, функции состояния · тепло и работа · первый закон · работа расширения: обратимые и необратимые пути · энтальпия и теплоемкость · закон Кирхгофа · бомбовый калориметр · почему H2 имеет 5/2 R при 300 K.

Young man: Sage, in the first book we constantly wrote ΔH, ΔG, ΔS as ready numbers. Let's finally understand what lies behind them.

Sage: Alright. Let's start with the very word. Thermodynamics answers the question "where to **and** how **far"** a process can go; kinetics — "how **fast." These** are different questions: a reaction can be favorable and still not occur (diamond to graphite) — we have seen this. Today we are only interested in the first.

System, environment, and state functions

Sage: Everything we study is called a **system;** everything else is **the environment.** A system can **be isolated** (exchanging neither energy nor matter), closed (exchanging only **energy),** and open (exchanging both energy **and matter).** Now for the main distinction. A quantity is a state **function if its** value depends only on the current state of the system and not on the path. Temperature, pressure, volume, internal energy U are state functions. Heat Q and work w are not.

A young man: Like a mountain: the height above sea level does not depend on the path, but the distance traveled does.

Sage: An exact image. This distinction will later give us Hess's law and everything else.

First principle

Sage: Formulation: ΔU = Q + **w,** where Q is the heat transferred to the system, w is the work done on the system (as accepted by IUPAC). Energy is neither created nor destroyed, but only changes from one form to another.

[UV] Energy conservation

The first law (Mayer, Joule, Helmholtz, 1840s) is confirmed by countless measurements; a perpetual motion machine of the first kind is impossible.

[OS] "Heat is contained in the body"

Heat is not a substance ("caloric") and not a reserve: it is a way of transferring energy due to temperature differences. Rumford (1798) noted that when drilling cannons, heat is "taken from nothing" — from work; Joule (1840s) measured the mechanical equivalent of heat. It is incorrect to speak of a "reserve of heat" in a body — one can speak of internal energy.

The work of expansion: the path is important.

Sage: Let's consider 1 mole of an ideal gas at 298 K, which expands from 10 bar to 1 bar. The work of expansion against the external pressure is equal to $w = -p_{внеш} \cdot \Delta V$ (the negative sign indicates that the system is giving off energy). We will calculate two paths. The first: the gas is released immediately against 1 bar.

Young man: Initial volume $V_1 = RT/p = 8.314 \cdot 298/106 = 2.48$ L; final $V_2 = 24.8$ L, $\Delta V = 22.3$ L. Work $w = -105\ \text{Pa} \cdot 22.3 \cdot 10{-}3\ \text{m}3 = -2.23$ kJ.

Sage: The second path: infinitely slowly, at each step the external pressure is just slightly less than the pressure of **the gas.** The work of reversible isothermal expansion: $w = -nRT \cdot \ln(V_2/V_1) = -nRT \cdot \ln(p_1/p_2)$.

Young man: $-2.479\ \text{kJ} \cdot \ln 10 = -2.479 \cdot 2.303 = -5.71$ kJ. 2.56 times more!

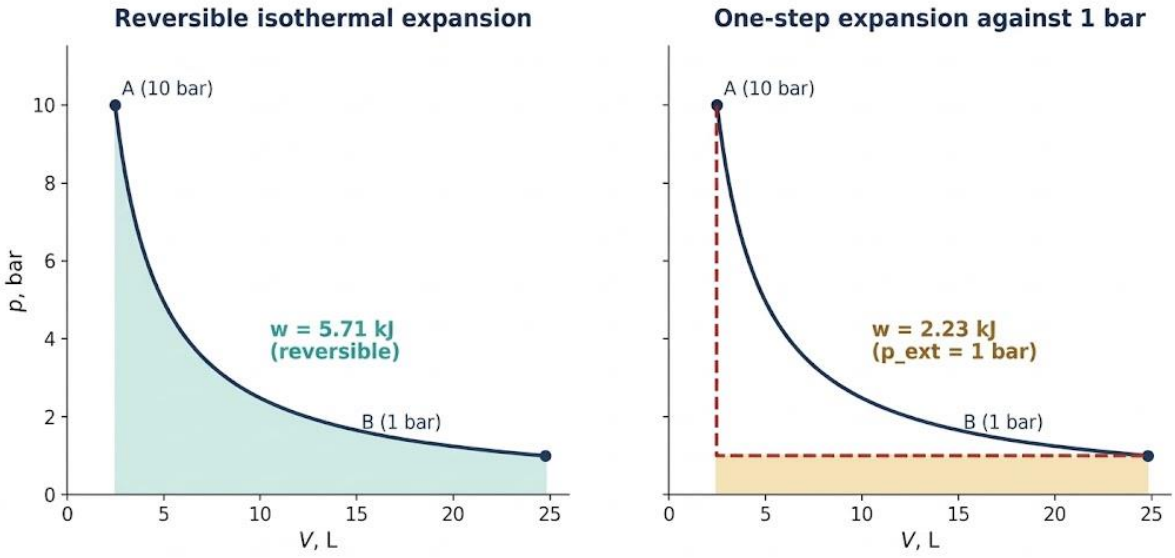


Fig. 1. Expansion work of an ideal gas: two paths between the same states.

Sage: How has the internal energy changed?

Young man: For an ideal gas, U depends only on T. T has not changed, so $\Delta U = 0$ in both cases. Then $Q = -w$: in the first case, the gas absorbed 2.23 kJ of heat from the surroundings, in the second — 5.71 kJ. Work and heat depend on the path, but ΔU does not!

> **[OD] Reversibility — idealization**
> A reversible process is the limit of infinitely slow changes, during which the system is almost in equilibrium at every moment. Real processes are irreversible; the reversible path provides an upper limit on the work obtained and allows for the calculation of state functions.

Enthalpy and heat capacity

Sage: Most chemical reactions occur at constant pressure. It is convenient to introduce enthalpy $H = U + \mathbf{pV}$. Then at constant p, heat is equal to the change in enthalpy: $Q_p = \Delta H$. For reactions with gases $\Delta H = \Delta U + \Delta n_r \cdot RT$. Let's take $H_2 + \frac{1}{2}O_2 \rightarrow H_2O(l)$: $\Delta H = -285.8$ kJ/mol.

Young man: $\Delta n_r = 0 - 1.5 = -1.5$. $\Delta U = \Delta H - \Delta nRT = -285.8 + 1.5 \cdot 2.479 = -285.8 + 3.72 = -282.1$ kJ/mol. The difference is 3.7 kJ.

Sage: Now the heat capacity: C = dQ/dT. At constant volume $C_V = (\partial U/\partial T)_V$, at constant pressure $C_p = (\partial H/\partial T)_p$. For an ideal gas $C_p - C_V = R$. Look at the table.

Table 3. Molar heat capacities of gases at 298 K.

Gas	Cp, J/(mol·K) / J/(mol·K)	$C_V = C_p - R$	CV/R
He, Ar	20,8	12,5	1,5
N_2, O_2, H_2	29,1	20,8	2,5
CO_2	37,1	28,8	3,5
$H_2O(g)$ / $H_2O(g)$	33,6	25,3	3,0

Young man: For helium, $C_V = 1.5R$: three translational degrees of freedom, ½R for each. For nitrogen, it's 2.5R: two rotations have been added. But for CO_2, it's already 3.5R. Where does that come from?

Sage: The theorem of equipartition: each "quadratic" degree of freedom gives ½R. A diatomic molecule, if classical mechanics were applicable, would have an additional vibration — that is +R, totaling 3.5R. But for nitrogen at 300 K, there are only 2.5R. Look at the figure.

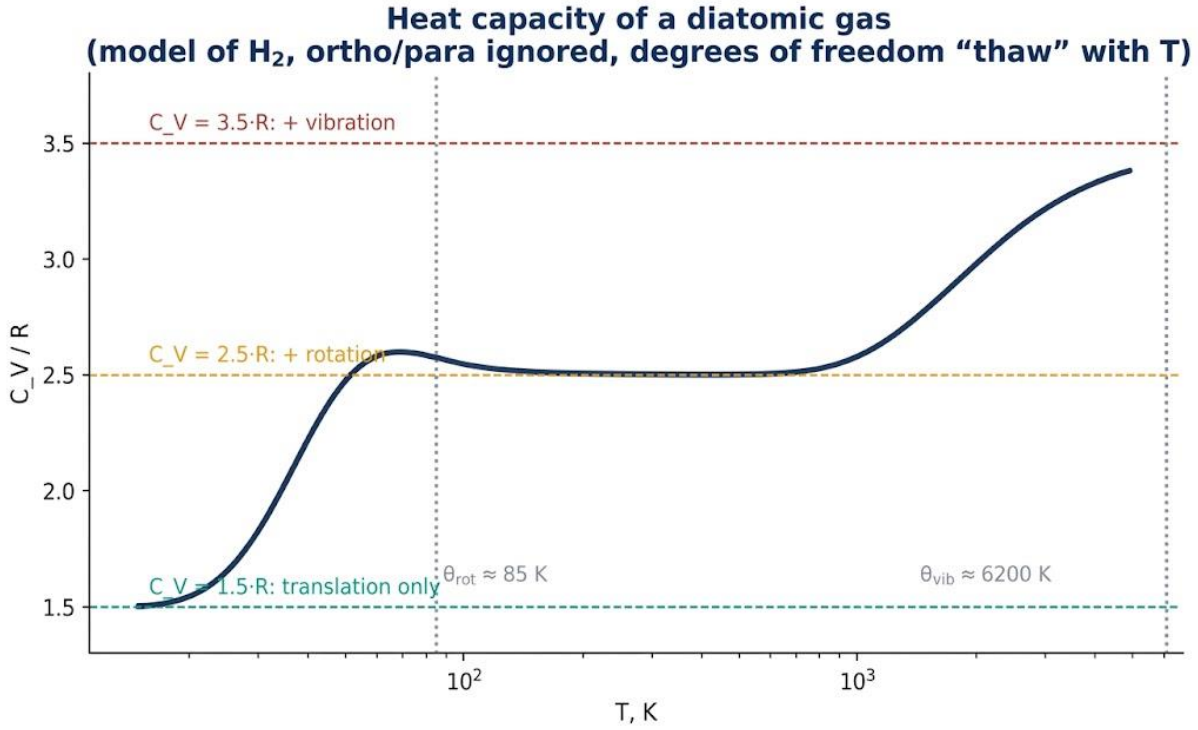


Fig. 2. Degrees of freedom “thaw” with temperature.

Young man: The fluctuation of the contribution does not occur until T becomes comparable to $\theta_{col} \approx 6200$ K. But why?

Sage: Because the energy of vibrations is quantized: the nearest level above the ground state is at ħω, which is 6200 K in temperature units. At 300 K, the molecules simply have nothing to "pay" for the first quantum. Classical physics did not explain this; Einstein applied the idea

of quanta to the heat capacity of solids in 1907, and this became one of the first successes of quantum theory.

> **[UV] Quantum nature of heat capacity**
> The stepwise increase in the heat capacity of a diatomic gas with temperature is observed experimentally and is explained by the quantization of rotational and vibrational levels. This is one of the threads that connects thermodynamics with quantum chemistry (Part IV).

Kirchhoff's law: ΔH depends on temperature.

Sage: Since substances have different heat capacities, the thermal effect depends on T: $\Delta H(T2) = \Delta H(T1) + \Delta Cp \cdot (T2 - T1)$ (at constant Cp). Applied to the synthesis of ammonia $N2 + 3H2 \rightarrow 2NH3$, $\Delta H(298) = -92.4$ kJ. Cp (J/(mol·K)): N2 29.1; H2 28.8; NH3 35.1.

Young man: $\Delta Cp = 2 \cdot 35.1 - 29.1 - 3 \cdot 28.8 = 70.2 - 29.1 - 86.4 = -45.3$ J/K. At 700 K: $\Delta H \approx {}^{-92.4} - 45.3 \cdot 402 \cdot 10{-}3 = -92.4 - 18.2 = -110.6$ kJ. The thermal effect has become significantly more negative.

> **[OD] Clarification to Chapter 8 of the First Book**
> There we evaluated the equilibrium constant at 450 °C, assuming ΔH to be constant. Kirchhoff's law shows that at high temperatures, |ΔH| is greater; the order of magnitude estimate does not change, but for precise calculations, a correction must be introduced.

Calorimetry: how to measure it

The sage: The combustion ΔU is **measured in a bomb calorimeter:** the substance is burned in a steel vessel of constant volume, immersed in water. The heat of the reaction goes to heating the water and the calorimeter. For 1.000 g of benzoic acid, the combustion ΔU is −26.4 kJ/g. Let the heat capacity of the calorimeter with water be 10.0 kJ/K.

Young man: $\Delta T = 26.4$ kJ / 10.0 kJ/K = 2.64 K. Such a temperature increase can be easily measured with a thermometer to an accuracy of thousandths of a degree. From this, ΔU is determined, and then it is converted to ΔH.

Chapter summary

- $\Delta U = Q + w$; U is a state function, Q and w are not.
- $Qp = \Delta H$; $\Delta H = \Delta U + \Delta nRT$; $Cp - CV = R$ (ideal gas).
- Heat capacity reflects quantized energy levels; ΔH depends on T (Kirchhoff's law).

Self-check

- 1 mol He is heated from 300 to 400 K at constant p. Find Q, ΔH, ΔU, w. (2.08; 2.08; 1.25; −0.83 kJ/kJ)
- Why is C_V/R of H_2O(g) 3 rather than 3.5? (non-linear molecule: 3 rotations)

Glossary

Russian	Английский
system, environment	система, окружающая среда

Russian	Английский
state function	функция состояния
internal energy	внутренняя энергия
enthalpy	энтальпия
heat capacity	теплоемкость
reversible process	обратимый процесс
calorimeter	calorimeter

Chapter 2. Entropy and the Second Law

In this chapter

Why another quantity is needed · Carnot cycle · entropy according to Clausius and Boltzmann · second law: entropy of the Universe · standard entropies of substances · third law and "Paulings ice" · entropy and information.

Почему нужна еще одна величина · цикл Карно · энтропия Клаузиуса и Больцмана · второй закон · стандартные энтропии · третий закон и лед Полига · энтропия и информация.

Young man: Sage, the first principle allows for any transformations of energy, as long as its sum is conserved. But hot tea always cools down, rather than heating up from the cold air — although energy would still be conserved in that case. What is wrong here?

Sage: The first principle is unaware of **direction. The** direction is set by the second. Historically, it arose from the practice of steam engines: Carnot (1824) discovered that a heat engine cannot convert all heat into work — part must be transferred to the refrigerator. The maximum efficiency of the machine between temperatures Th and Tc: η = **1 − Tc/Th.**

Young man: For 500 K and 300 K, this is 1 − 0.6 = 0.4: 40%. The rest is inevitably given to the refrigerator.

The sage: Clausius (1865) saw that there is a new state function behind **this** — entropy, defined for a reversible process: **dS** = δQобр/T. And the second law states: in any real process in an isolated system, entropy does not decrease.

Calculating entropy

Sage: Ice melts at 273.15 K, absorbing ΔHпл = 6.01 kJ/mol. The phase transition at constant T is reversible, thus ΔS = Q/T.

Young man: ΔS = 6010/273.15 = 22.0 J/(mol·K). And the boiling of water: 40,660 J/373.15 K = 109 J/(mol·K).

The sage: Boiling provides a much greater jump: gas offers incomparably more ways to arrange molecules. For many liquids, ΔS_kip ≈ 88 J/(mol·K) — the Trouton's rule. Water has more due to hydrogen bonds: the liquid is more ordered.

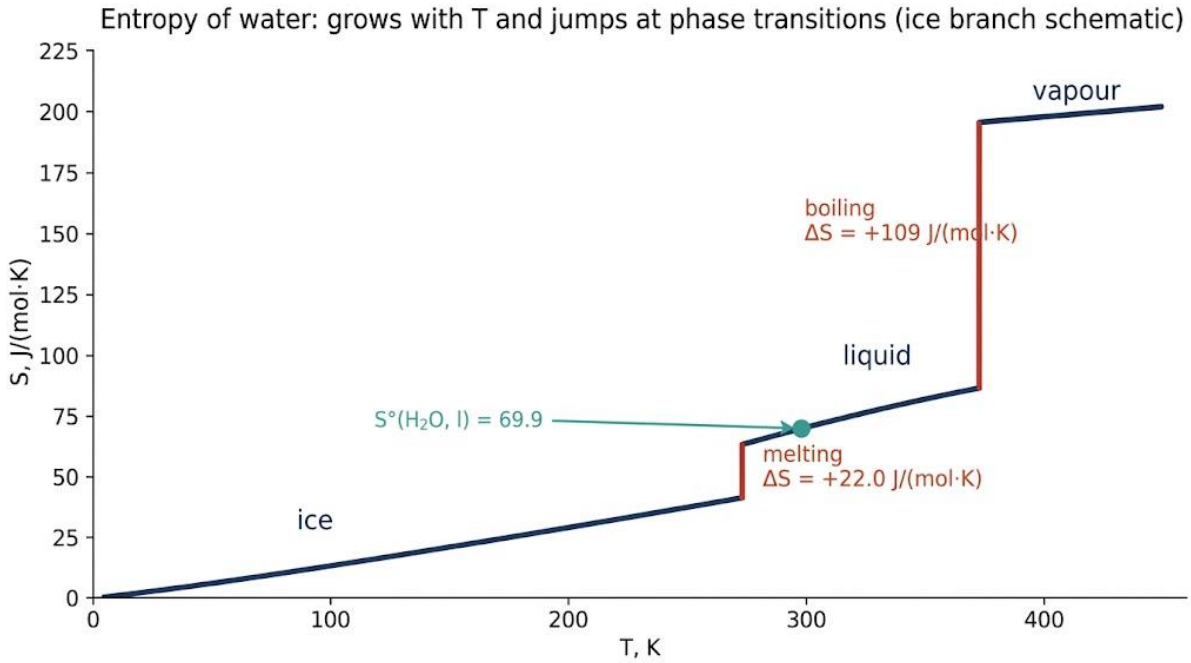


Fig. 3. Entropy of water: growth with temperature and jumps at phase transitions.

Boltzmann entropy: counting the ways

The sage: Boltzmann (1877) gave entropy a molecular meaning: S = **kB·ln W,** where W is the number of microstates realizing a given macrostate. Let the gas expand twice: each of the N molecules received twice as many places.

The young man: W increased by 2N times. ΔS = kB·ln 2N = N·kB·ln 2 = nR·ln 2 = 8.314·0.693 = 5.76 J/K per mole. This is the same number given by Clausius's formula for reversible expansion!

Sage: Now it is clear why gas never collects on one side of the vessel by itself. Count the ways for 10 molecules.

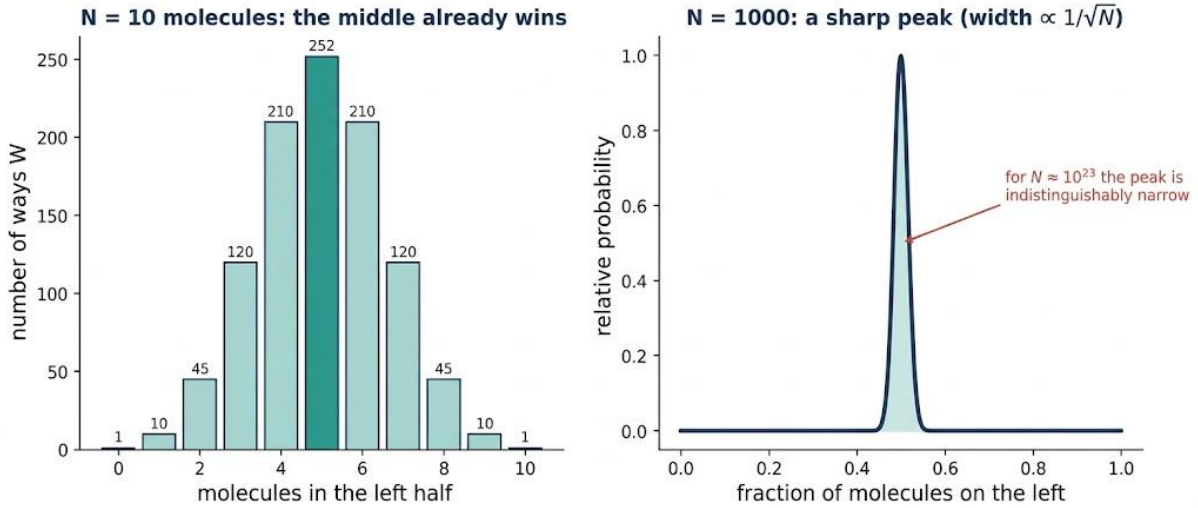


Fig. 4. The statistical meaning of the second law: counting arrangements.

Young man: All 10 on the left — one configuration out of 1024; evenly 5 and 5 — 252 out of 1024. For N = 1000, the peak is already narrow, and for 1023 molecules, it is infinitely sharp. Therefore, "impossible" actually means "inconceivably unlikely."

[UV] The second law - a statistical law
For macroscopic systems, deviations from the second law are so incredible that they are never observed. For small systems (a few molecules), fluctuations are noticeable, and this has been measured.

[OD] "Entropy is disorder"
The term "image" can be considered as a first approximation, but it is better to say "the number of available microstates." There are known cases where a more "ordered" state has greater entropy: for example, at sufficiently high density, solid spheres crystallize precisely because, in an ordered packing, each has more space to move (Alder and Wainwright, 1957). Therefore, the word "disorder" should be used with caution.

Entropy of the Universe

Sage: For an isolated system ΔS ≥ 0. And what about the tea in the room? Let 1000 J transfer from the hot body (400 K) to the cold one (300 K).

Young man: The hot one loses: ΔS1 = −1000/400 = −2.5 J/K. The cold one gains: ΔS2 = +1000/300 = +3.33 J/K. The sum: +0.83 J/K. Positive!

Sage: That is why heat flows from hot to cold, and not the other way around. And notice, nothing is **"violated":** the entropy of one body can decrease (the hot one) if the total increases. The Earth is an open system: a living organism maintains order by increasing the entropy of the environment (this is how plants operate, receiving low-entropy sunlight and releasing high-entropy heat).

[OS] "Life Violates the Second Law"
The second law refers to **isolated systems** or the sum of "system + environment." Living organisms are open systems, and their order is paid for by the increase in entropy of the environment.

[Γ] Where does the low initial entropy come from?
In order for the entropy of the Universe to increase, it was very low in the past. The reason for this is not established; a popular explanation is related to the initial conditions of the cosmological model ("past hypothesis"). This is a hypothesis, not a fact.

Standard entropies and the third law

Sage: Entropy can be calculated using heat capacity: $S(T) = \int (C_p/T)dT$. $_{Zero}$ gives the third **law: the** entropy of an ideal crystal at 0 K is equal to zero (Nernst, Planck). Therefore, unlike H, there is an absolute scale for S.

Table 4. Standard molar entropies.

Substance	S°(298 K), J/(mol·K) / J/(mol·K)
C (diamond)	2,4
C (graphite)	5,7

Substance	S°(298 K), J/(mol·K) / J/(mol·K)
Fe(тв) / Fe(s)	27,3
H2O(j) / H2O(l)	69,9
NaCl(tv) / NaCl(s)	72,1
He(g) / He(g)	126,2
H2(g) / H2(g)	130,7
CH4(g) / CH4(g)	186,3
H2O(g) / H2O(g)	188,8
N2(g) / N2(g)	191,6
NH3(g) / NH3(g)	192,8
O2(g) / O2(g)	205,2
CO2(g) / CO2(g)	213,8

Young man: The pattern is clear: solids are few, liquids are more, gases are the most. And diamond with its rigid lattice is less than graphite. And entropy increases with mass and complexity of the molecule.

Sage: Let's apply it to the synthesis of ammonia: $N_2 + 3H_2 \rightarrow {}_{2NH_3}$.

Young man: $\Delta S^\circ = 2\cdot192.8 - 191.6 - 3\cdot130.7 = 385.6 - 191.6 - 392.1 = -198.1$ J/K. From four moles of gas, two were obtained: the entropy dropped sharply.

Sage: In chapter 8 of the first book, we mentioned that the yield of ammonia decreases upon heating. Now it is clear why: $\Delta H < 0$ "for" the reaction, but $\Delta S < 0$ "against," and with an increase in T, the contribution of $-T\Delta S$ grows (chapter 3).

Sage: One caveat about "zero" entropy. For ice, it is not equal to zero at 0 K: it has been measured at 3.41 J/(mol·K). Pauling explained this: hydrogen bonds in ice can be arranged in several ways, and about 3/2 ways per molecule are "frozen."

Young man: $R\cdot\ln(3/2) = 8.314\cdot0.405 = 3.37$ J/(mol·K). Wow, it matched the measurement!

> **[UV] Residual entropy of ice**
> The measured value of 3.41 J/(mol·K) and the Pauling estimate of 3.37 J/(mol·K) agree with an accuracy of about 1%. This is an elegant confirmation of the relationship $S = k\cdot\ln W$.

Entropy and information

The sage: Erasing one bit of information in an environment at temperature T inevitably leads to the release of heat of no less than $kBT\cdot\ln 2$ (Landauer's principle): at 300 K this is $2.9\cdot10{-}21$ J. This result resolved the famous paradox of "Maxwell's demon" and was experimentally verified in 2012.

> **[UF] Landauer's Principle**
> The lower bound of heat during the erasure of a bit $kBT\cdot\ln 2$ has been experimentally confirmed on a colloidal particle in an optical trap (Beryu et al., 2012). The connection between entropy and information is not a metaphor.

Chapter summary

- The second law gives direction: ΔS(universe) ≥ 0.
- dS = δQобр/T (Clausius); S = kB·ln W (Boltzmann). / dS = δQrev/T (Clausius); S = kB·ln W (Boltzmann).
- Absolute entropies exist thanks to the third law; ice retains 3.4 J/(mol·K).

Self-check

- Calculate ΔS° for CaCO3 → CaO + CO_2, if S°: CaCO3 92.9; CaO 39.8; CO2 213.8. / Compute ΔS° for CaCO3 → CaO + CO2. (+160.7 J/K)
- What is the maximum efficiency of a heat engine between 800 K and 300 K? (62.5 %)

Glossary

Russian	Английский
entropy	энтропия
the second law of thermodynamics	второй закон термодинамики
third law	третья закон
microstate	микросостояние
Carnot cycle	Карно́тов цикл
residual entropy	остаточная энтропия

Chapter 3. Free Energy, Chemical Potential and Equilibrium

In this chapter

The output ΔG = ΔH − TΔS · chemical potential · ΔG° and equilibrium constant · ΔG = ΔG° + RT·ln Q · dependence of K on temperature (van 't Hoff equation) · decomposition CaCO3 · maximum work and efficiency of the fuel cell.

Derivation of ΔG = ΔH − TΔS · chemical potential · ΔG° and the equilibrium constant · ΔG = ΔG° + RT·ln Q · temperature dependence of K (van't Hoff) · CaCO3 decomposition · maximum work and fuel-cell efficiency.

Young man: In the first book, we simply stated: "the reaction occurs if ΔG < 0". I want to understand where this comes from.

Sage: We will derive it in three lines. The second law: $\Delta S_{Universe} = \Delta S_{system} + \Delta S_{surroundings} \geq 0$. If the reaction occurs at constant T and p, then the surroundings receive heat −ΔH (what the system has given off), hence $\Delta S_{surroundings} = -\Delta H/T$. Let's substitute.

Young man: $\Delta S_{universe} = \Delta S - \Delta H/T \geq 0$. Multiply by −T: ΔH − TΔS ≤ 0. This is ΔG ≤ 0, where G = **H** − TS — Gibbs energy.

Sage: Correct. Free energy is merely the second law rewritten for constant T and p, in terms of just one system. The minimum G indicates equilibrium; ΔG < 0 indicates spontaneity. Note also: |ΔG| is the **maximum** useful (non-expansion) work that can be obtained: $w_{max} = \Delta G$.

Chemical potential

Sage: For a mixture, it is convenient **to introduce the chemical potential** $\mu_i = (\partial G/\partial n_i)_{T,p}$: how much G will change with the addition of one mole of substance i. For an ideal gas $\mu = \mu^\circ + RT\ln(p/p^\circ)$; for a solution $\mu = \mu^\circ + RT\ln a$ (a — activity). The equilibrium condition: $\sum \nu_i \mu_i = 0$ — the potentials of the products and reactants are balanced.

Sage: From this follows the central formula. For the reaction ΔG = ΔG° + RT·ln Q, where Q is the reaction quotient (the formula is the same as for K, but for the current concentrations). At equilibrium ΔG = 0 and Q = K, from which **$\Delta G^\circ = -RT\ln K$.**

[UV] Direction of the reaction

If Q < K, then ΔG < 0 and the reaction proceeds forward; if Q > K, it goes backward; if Q = K — equilibrium. Le Chatelier's principle is a consequence of this statement.

Table 5. Relationship between ΔG° and K at 298 K.

ΔG°, kJ/mol	K (298 K)	What does this mean / Meaning
−40	$1{,}0\cdot10^{7}$	goes to completion
−20	$3{,}2\cdot10^{3}$	products are much more numerous
−10	57	products prevail
0	1	a mixture of both / balanced
+10	0,018	reactants prevail

ΔG°, kJ/mol	K (298 K)	What does this mean / Meaning
+20	$3{,}1\cdot10^{-4}$	hardly proceeds
+40	$1{,}0\cdot10^{-7}$	negligible

Young man: Every 5.7 kJ (that is, RT·ln 10) changes K by ten times. A good scale in the head!

Example: $N_2O_4 \rightleftharpoons 2NO_2$

Sage: This is a beautiful reversible reaction: N_2O_4 is colorless, NO_2 is brown. Data at 298 K: $\Delta G_{rev}(N_2O_4) = 97.9$ kJ, $\Delta G_{rev}(NO_2) = 51.3$ kJ; ΔH_{rev}: 9.2 and 33.2 kJ. Find K and the degree of dissociation at 1 bar.

Young man: $\Delta G^\circ = 2\cdot51.3 - 97.9 = 4.7$ kJ. $K = \exp(-4700/2479) = e^{-1.90} = 0.150$. For α: $K = 4\alpha^2/(1-\alpha^2)$ at p = 1 bar, $\alpha = \sqrt{(K/(K+4))} = \sqrt{(0.150/4.150)} = 0.19$. 19% dissociated.

Sage: Now heat to 373 K. $\Delta H^\circ = 2\cdot33.2 - 9.2 = 57.2$ kJ. Van 't Hoff equation: $\ln(K_2/K_1) = (\Delta H^\circ/R)\cdot(1/T_1 - 1/T_2)$.

Young man: $\ln(K_2/K_1) = (57\ 200/8{,}314)\cdot(1/298 - 1/373) = 6880\cdot0{,}000675 = 4.64$. $K_2 = 0.150\cdot e^{4.64} = 0.150\cdot104 = 15.6$. $\alpha = \sqrt{(15.6/19.6)} = 0.89$.

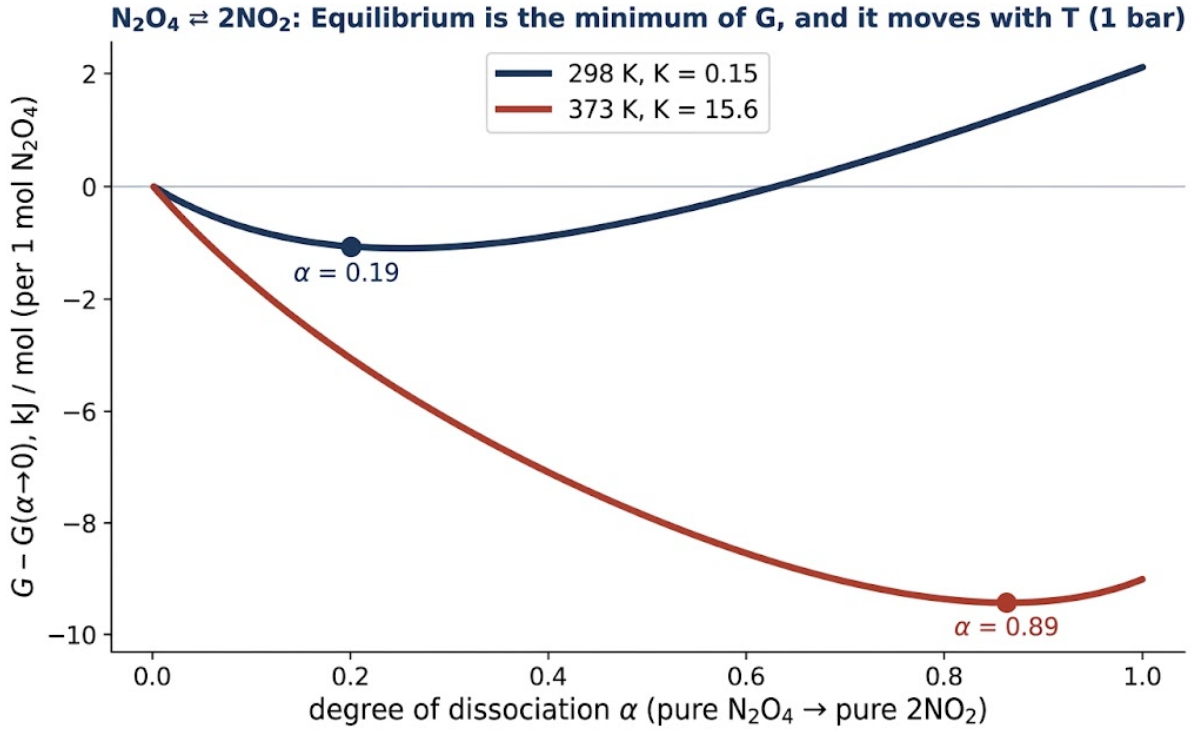


Fig. 5. Equilibrium is a minimum of G whose position depends on T.

[UV] Color shift
The flask with NO_2/N_2O_4 darkens upon heating (more brown NO_2) and lightens upon cooling. The calculation (19 % → 89 %) agrees well with the observation.

Temperature and ΔG: decomposition of calcium carbonate

Sage: ΔG = ΔH − TΔS. If ΔH and ΔS have the same sign, there exists a temperature T = ΔH/ΔS at which ΔG = 0. For CaCO3 → CaO + CO2: ΔH = +178.3 kJ, ΔS = +160.5 J/K.

Young man: T = 178 300/160.5 = 1111 K, that is, 838 °C, at standard pressure CO2 1 bar. Industrial firing is conducted at around 900 °C — slightly above this limit, as it should be.

Sage: And if CO2 is constantly being carried away from the furnace? Then Q = p(CO2)/p° is small. Look at how the equilibrium temperature changes.

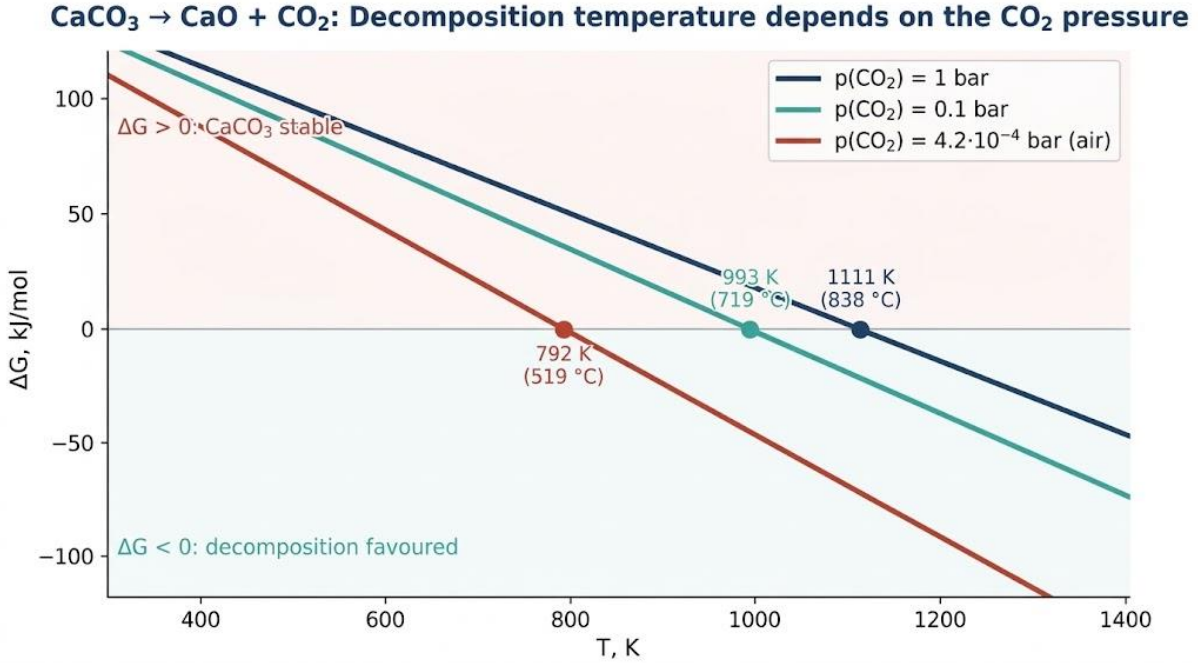


Fig. 6. ΔG of CaCO3 decomposition at different CO2 pressures.

Young man: At p(CO2) = 0.1 bar Tравн = 993 K (720 °C), and in air (4.2·10−4 bar) — 792 K (519 °C). The lower CO2 around, the easier the decomposition. But practically, higher temperatures are needed.

The sage: Indeed, thermodynamics says that it is possible, **while** the rate is determined by kinetics (Part III). But the principle of "removing the product shifts the equilibrium" is the same as that used in Chapter 8 of the first book.

> **[OD] Evaluation at constant ΔH and ΔS**
> The values 1111, 993, and 792 K were obtained under the approximation of constant ΔH and ΔS. Taking into account heat capacities (Kirchhoff's law, chapter 1) gives small corrections. The actual firing temperatures in industry are higher due to the speed of the process.

Sage: And one more important conclusion for the synthesis of ammonia: ΔH = −92.2 kJ, ΔS = −198.1 J/K.

Young man: ΔG = 0 at T = ΔH/ΔS = 92 200/198.1 = 465 K, that is, 192 °C. Above this temperature ΔG° > 0, and at standard pressure of 1 bar, the decomposition $_{NH_3}$ is favored, not the formation. That is why high pressure is needed!

Maximum work: fuel cell versus heat engine

Sage: We will burn hydrogen: $H_2 + \tfrac{1}{2}O_2 \rightarrow H_2O(l)$. $\Delta H = -285.8$ kJ, $\Delta G = -237.1$ kJ. In a conventional heat engine, we obtain only a fraction of the heat ΔH, limited by the Carnot efficiency. In a fuel cell, the reaction occurs electrochemically, and the maximum electrical work is equal to ΔG.

Young man: The limiting ratio: $\Delta G/\Delta H = 237.1/285.8 = 0.83$. And for the heat engine between 800 and 300 K according to Carnot 62.5%.

> **[OD] Ideal limits**
> The numbers 83% and 62.5% are theoretical upper limits. Real fuel cells operate with an efficiency of about 40–60%, while thermal machines operate below the Carnot limit. The advantage of the fuel cell is that it is not constrained by the Carnot cycle; more details are in Chapter 6.

Chapter summary

- $\Delta G = \Delta H - T\Delta S$ is the second law for constant T and p; $|\Delta G|$ is the maximum useful work.
- $\Delta G = \Delta G^\circ + RT\cdot\ln Q$; $\Delta G^\circ = -RT\cdot\ln K$. / $\Delta G = \Delta G^\circ + RT\cdot\ln Q$; $\Delta G^\circ = -RT\cdot\ln K$.
- The van’t Hoff equation gives K(T); temperature shifts the minimum of G.

Self-check

- Find K at 298 K for $\Delta G^\circ = -29$ kJ/mol. (≈ 1.2·10^5)
- Limestone calcination in a draughty kiln proceeds at a lower temperature than in a closed vessel due to the continuous removal of CO_2 gas produced during the reaction. In a closed vessel, the accumulation of CO_2 can inhibit the reaction, requiring higher temperatures to drive the process forward. The draught in the kiln facilitates the escape of CO_2, allowing the reaction to proceed more efficiently at lower temperatures.

Glossary

Russian	Английский
Gibbs energy	Энергия Гиббса
chemical potential	химический потенциал
reaction coefficient	реакционный коэффициент
equilibrium constant	константа равновесия
Van 't Hoff equation	уравнение ван't Гоффа
maximum work	максимальная работа

Chapter 4. Phases and Solutions

In this chapter

Phases, components, phase rule · Clapeyron and Clausius–Clapeyron equations · phase diagram of water · ideal solutions and Raoult's law · mixing energy · colligative properties · Henry's law.

Фазы, компоненты, фазовое правило · уравнения Клапейрона и Клаузиуса–Клапейрона · фазовая диаграмма воды · идеальные растворы и закон Рауля · свободная энергия смешивания · коллигативные свойства · закон Генри.

Young man: Sage, why does water boil at 100 °C — and why does it boil earlier in the mountains?

Sage: Because boiling is a phase equilibrium between liquid and vapor, and its temperature depends on pressure. A phase is a homogeneous part **of** the system, separated by a boundary; a component is a independently **variable substance.** Gibbs derived the phase rule in 1876: the number of degrees of freedom **F** = C − P + 2, where C is the number of components, P is the number of phases.

Young man: For pure water C = 1. In the single-phase region (P = 1) F = 2: T and p can be changed independently. On the line (P = 2) F = 1: if T is set, then p is determined. At the triple point (P = 3) F = 0: it is unique.

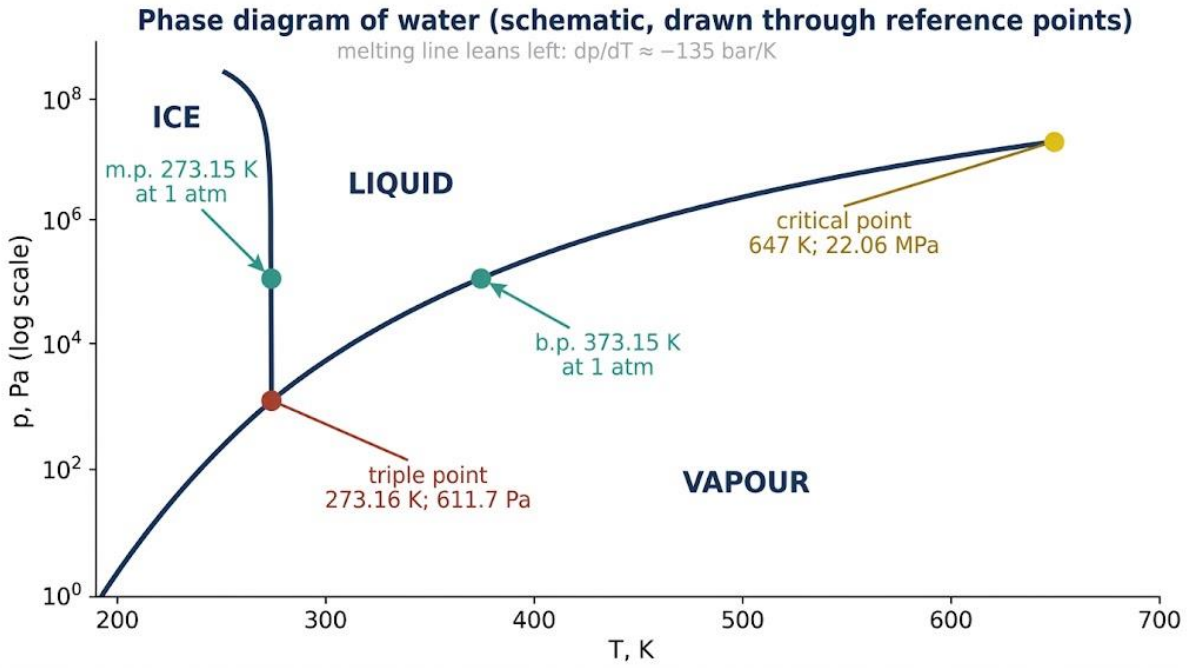


Fig. 7. The phase diagram of water.

Clausius and Clausius-Clapeyron

The slope of the equilibrium line is given by the Clapeyron equation: dp/dT = ΔH/(T·ΔV). For the melting of ice, ΔH = 6010 J/mol, **and the** volume decreases: ΔV = 18.02/0.9998 − 18.02/0.917 = −1.63 cm³/mol.

Young man: dp/dT = 6010/(273·(−1.63·10−6)) = −1.35·10^7 Pa/K = −135 bar/K. The melting line is sloped to the left: an increase in pressure of 135 bar lowers the melting temperature of ice by 1 K.

> **[OS] Do skates glide because the ice melts under the blade?**
> The calculation shows that the pressure under the blade reduces the melting temperature by only a fraction of a degree, and ice slides even at −20 °C. Explanations (friction, pre-melting of the surface layer) are still being discussed; the mechanism is not fully established.

Sage: For evaporation ΔV ≈ V_vapor (liquid can be neglected), and vapor can be considered ideal: V = RT/p. We obtain the Clausius–Clapeyron equation: ln(p2/p1) = −(ΔH_vapor/R)·(1/T2 − 1/T1). ΔH_vapor of water is 40.66 kJ/mol. At an altitude of 3000 m, the pressure is about 70 kPa. Find the boiling point t.

Young man: ln(70/101.3) = −0.370. Then 1/T2 − 1/373.15 = 0.370/(40660/8.314) = 0.370/4891 = 7.57·10−5. 1/T2 = 0.002680 + 0.0000757 = 0.002756, T2 = 362.9 K ≈ 90 °C. And in a pressure cooker at 2 bar?

Sage: Count.

Young man: ln 2 = 0.693; 1/T2 = 0.002680 − 0.693/4891 = 0.002680 − 0.0001417 = 0.002538. T2 = 394 K ≈ 121 °C. Therefore, everything cooks faster in a pressure cooker: for every 10 degrees, reactions speed up by about twice (this is discussed in Part III).

Ideal solutions and mixing energy

Sage: In an ideal solution, the molecules of the components interact with each other just as they do among themselves. Raoult's law: the vapor pressure of component A over the solution pA = xA·pA*. The enthalpy of mixing is zero, and the Gibbs **energy: ΔGсмеш = nRT·Σxi·ln xi.**

Young man: We mix 1 mole of A and B at 298 K: x = 0.5 of each. ΔG = 2·2.479·(0.5·ln0.5 + 0.5·ln0.5) = 2·2.479·(−0.693) = −3.44 kJ. It is always negative: ideal gases and solutions mix spontaneously only due to entropy.

Sage: And if the molecules are not identical and "do not like" each other? The interaction parameter χ is introduced: ΔGсмеш/(nRT) = x·ln x + (1 − x)·ln(1 − x) + χ·x(1 − x). Look at what χ does.

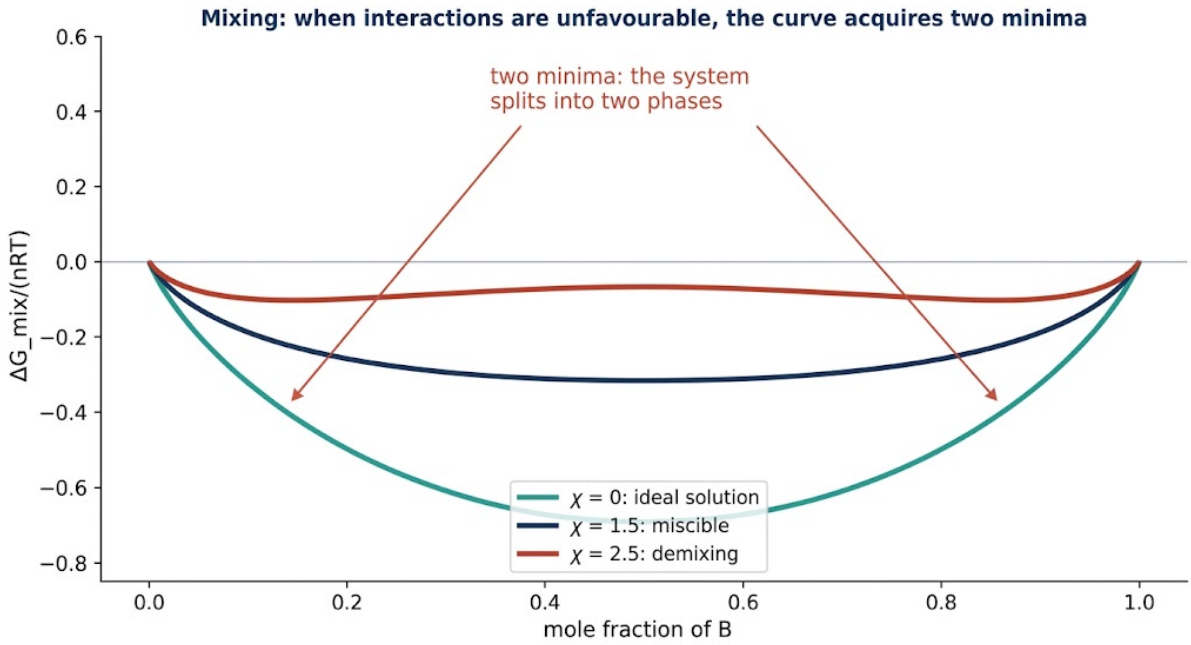


Fig. 8. Free energy of mixing for different interactions.

Young man: At χ = 2.5, the curve has two minima: the system is more advantageous to separate into two phases with compositions at the minima. This is how oil and water separate.

> **[OD] Regular solution model**
> The formula with the parameter χ is the simplest model (regular solution). It qualitatively describes phase separation, but real mixtures are more complex; the parameter χ often depends on T and composition.

Colligative properties

The properties that depend only on the **number** of dissolved particles, and not on their nature, are called colligative. A dissolved non-volatile substance lowers the vapor pressure and therefore raises the boiling point and lowers the freezing point: ΔTк = Kэ·m, ΔTз = Kк·m, where m is the molality, i is the number of particles (for an electrolyte).

Table 6. Ebullioscopic and cryoscopic constants.

Solvent	Kэ, K·kg/mol / Kb	Kк, K·kg/mol / Kf
water	0,512	1,86
benzene	2,53	5,12
acetic acid	3,07	3,90
camphor	—	≈ 37,7

Young man: A solution of 1 mole NaCl in 1 kg of water: i ≈ 2, ΔTз ≈ 2·1.86 = 3.7 K. In reality, about 3.4 K — ions interact (chapter 5).

Sage: Another colligative property is osmotic pressure Π = **c·R·T (c is** the total molar concentration of particles). For a physiological solution of 0.154 M NaCl, where the number of particles is almost twice as much.

Young man: c ≈ 0.154·2 = 0.308 mol/L = 308 mol/m³. Π = 308·8.314·310 = 7.9·10⁵ Pa ≈ 7.9 bar. The same as that of blood plasma!

> **[UV] Isotonic solution**
> The osmotic pressure of blood plasma is about 7–8 bar; a solution with the same pressure is called isotonic. Cells in it do not shrink or swell.

Gases in liquids and azeotrope

Sage: The solubility of a gas in a liquid is proportional to its partial pressure: Henry's **law, c** = kH·p. For oxygen above water at 25 °C and 1 atm O2, the solubility is about 1.3 mmol/L; from air (0.21 atm) about 0.27 mmol/L, which is 8 mg/L. The dissolution of a gas releases heat, so there is less oxygen in warm water.

Sage: For the ethanol-water mixture, the vapor-liquid curve has an azeotrope: **at** 95.6% by mass of ethanol and 78.2 °C, the composition of the vapor coincides with the composition of the liquid. Therefore, simple distillation cannot yield ethanol purer than 95.6%. Finally, above the critical point (for CO2 31 °C and 74 bar), the distinction between liquid and gas disappears — this is a supercritical fluid, a solvent for extraction.

> **[UV] Supercritical CO2**
> Above 31.0 °C and 73.8 bar CO2 — supercritical fluid: density is close to that of a liquid, viscosity is close to that of a gas. It is used, for example, for caffeine extraction. The equations in this chapter describe such a system only qualitatively; real equations of state are needed.

Chapter summary

- Phase rule F = C − P + 2; the Clapeyron equation sets the slopes of phase boundaries.
- Ideal solutions mix because of entropy; unfavourable interactions can cause demixing.
- Colligative properties depend on the number of particles: ΔTк, ΔTз, Π.

Self-check

- The freezing point of water will decrease by 0.93 K when 0.5 mol of glucose is dissolved in 1 kg of water.
- Estimate the boiling point of water at 0.5 atm (ΔH_vap = 40.7 kJ/mol).

Glossary

Russian	Английский
phase, component	фаза, компонент
phase rule	фазовое правило
triple point	тройная точка
Raoult's law	Рауля закон
colligative properties	colligative properties
osmotic pressure	осмотическое давление
azeotrope	azeotrope

Part I summary

We have determined that thermodynamics is based on three simple ideas. Energy is conserved (the first law); the entropy of the universe increases (the second); therefore, in a world with constant T and p, everything is determined by the minimum of G. On this basis, we obtained the equilibrium constant, phase diagrams, and solution properties. Now we can move on to reactions involving charged particles: electrochemistry.

What next / Next

Part II. Electrochemistry (chapters 5–6): ions in solution and electrical conductivity; electrode processes, fuel cells, and batteries.

Part II: Electrochemistry (chapters 5–6): ions in solution and conductivity; electrode processes, fuel cells and batteries.

PART II. Electrochemistry

Часть II. Электрохимия

Chapters 5–6

Chapter 5. Ions in Solution: Conductivity and Activity

In this chapter

How ions conduct current · molar conductivity and Kohlrausch's law · anomalous mobility of H+ (Grotthuss mechanism) · transport numbers · degree of dissociation by conductivity · ionic atmosphere and Debye–Hückel theory · activity and pH.

How ions carry current · molar conductivity and Kohlrausch's law · the anomalous mobility of H+ (Grotthuss mechanism) · transport numbers · degree of dissociation from conductivity · the ionic atmosphere and Debye–Hückel theory · activity and pH.

Young man: Sage, in the first book we wrote [H+] and [OH−] as if the ions in the solution are independent billiard balls. Is that so?

Sage: No, and today we will see how much there is none. Let's start with movement. If two electrodes are immersed in a solution, ions will flow: cations to the cathode, anions to the anode. The solution conducts current, and we measure the specific conductivity κ (S/cm). To compare substances, it is divided **by concentration: molar conductivity Λ** = 1000·κ/c (c in mol/L; Λ in S·cm²/mol).

The sage: Kohlrausch (1870s) found that for strong electrolytes Λ decreases linearly with √c: $\Lambda = \Lambda^\circ - K\cdot\sqrt{c}$. At infinite dilution, ions do not interfere with each other, and the law of **independent migration applies:** $\Lambda^\circ = \lambda^\circ_+ + \lambda^\circ_-$.

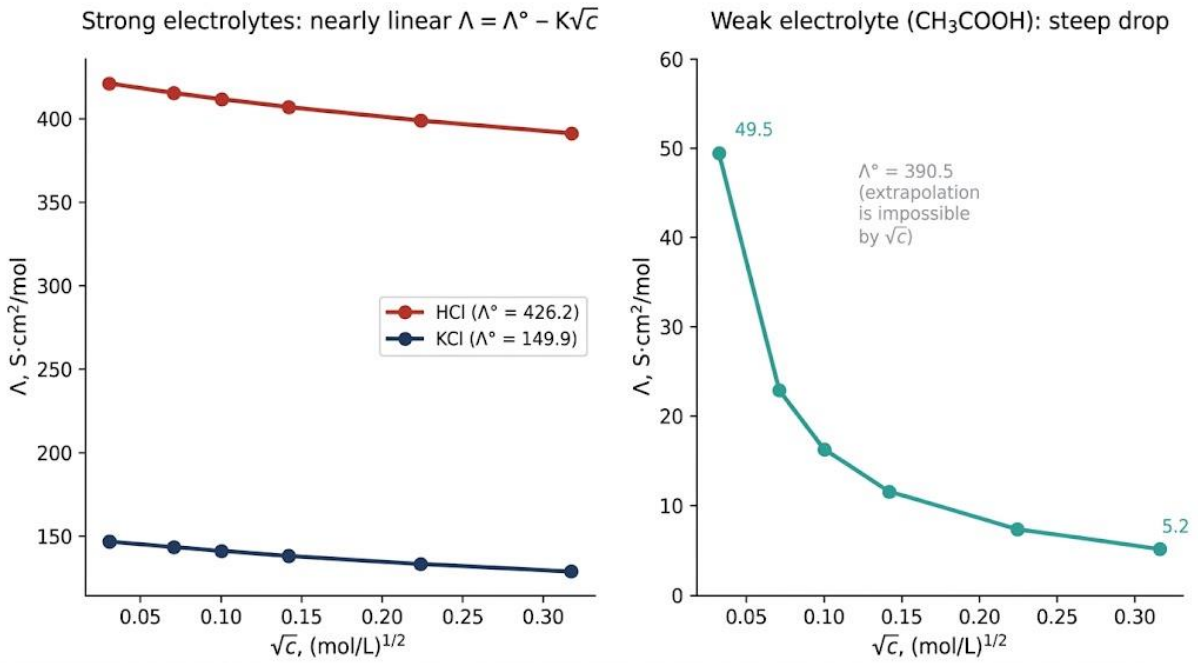


Fig. 9. Molar conductivity: strong and weak electrolytes.

Table 7. Limiting ionic conductivities in water at 25 °C.

Ion	λ°, Sm·cm²/mol / S·cm²/mol	Ion	λ°, Sm·cm²/mol / S·cm²/mol
H+	349,8	Cl−	76,4
Li+	38,7	Br−	78,1

Ion	λ°, Sm·cm²/mol / S·cm²/mol	Ion	λ°, Sm·cm²/mol / S·cm²/mol
Na+	50,1	NO3−	71,5
K+	73,5	CH3COO−	40,9
Ca2 + (½)	59,5	SO42− (½)	80,0
OH−	198,6		

Young man: I will check the law. Λ°(KCl) = 73.5 + 76.4 = 149.9, as shown in the graph. And Λ°(HCl) = 349.8 + 76.4 = 426.2. Wow, the mobility of H+ is seven times greater than that of Na+! The proton is lighter, after all.

Sage: It's not about ease. A regular ion carries a hydration shell around it and moves like a ball in a viscous liquid. A proton, on the other hand, is transferred in a relay: H3O+ donates a proton to a neighboring water molecule, which then donates it to the next one. The "carrier" itself hardly shifts, while the charge travels through a chain of hydrogen bonds. This is the Grotthuss mechanism (1806), and modern calculations and experiments have refined it (structural diffusion).

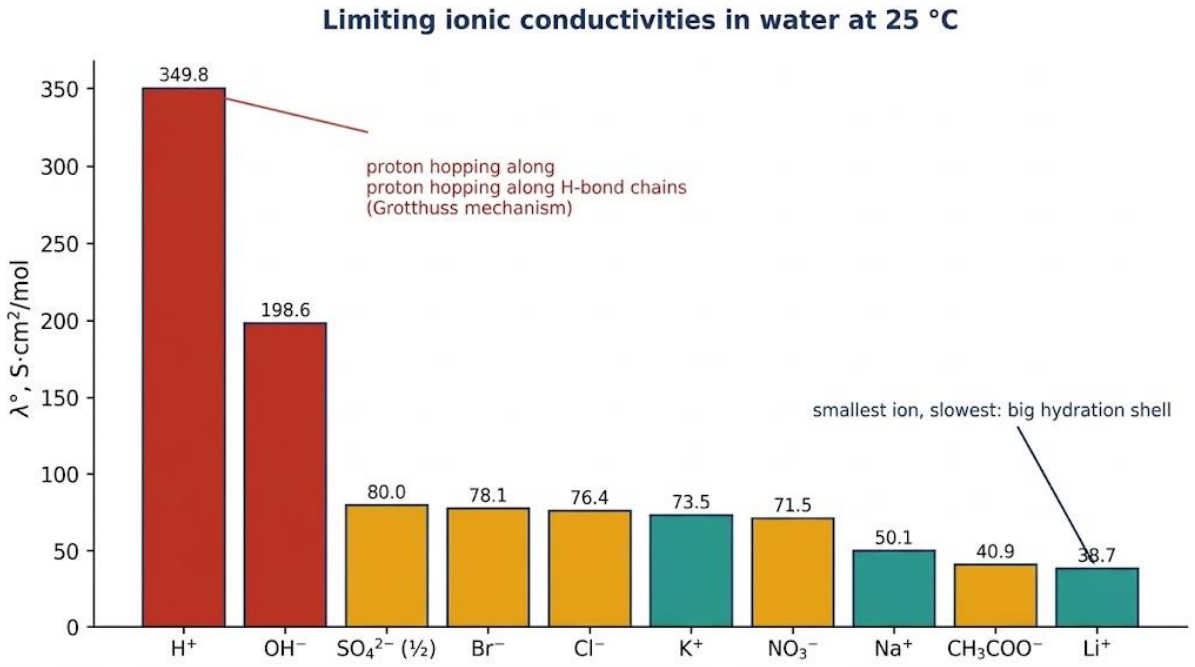


Fig. 10. Limiting ionic conductivities.

Young man: But lithium has the smallest mobility, even though the ion is the smallest! Strange.

Sage: Precisely because it is the smallest: it has the strongest electric field around it and the largest hydration shell. The effective size of an ion in solution is not the size in the crystal. This is the same "hydration paradox" from the energetics of dissolution (Chapter 9 of the first book).

[UV] Mobility of H+ and OH−
The abnormally high mobilities of H+ and OH− are explained by the proton hopping through the hydrogen bond network of water (Grotthuss mechanism). For other ions in water, "normal" viscous migration occurs, and mobility decreases with the increase of the hydration shell.

Transfer numbers and weak electrolytes

Sage: What fraction of the current is carried by the cation? The transport number $t_+ = \lambda+/(\lambda+ + \lambda-)$.

Young man: In HCl: 349.8/426.2 = 0.82. In KCl: 73.5/149.9 = 0.49. In NaCl: 50.1/126.5 = 0.40. In the hydrochloric acid solution, 82% of the current is carried by the proton!

Sage: How can the degree of dissociation of a weak acid be measured? The graph in √c is not suitable for it: Λ° cannot be obtained by extrapolation. However, it can be calculated using the law of independent migration.

Young man: $\Lambda°(CH_3COOH) = \lambda°(H+) + \lambda°(CH_3COO-) = 349.8 + 40.9 = 390.7$. At 0.1 M, Λ was measured as 5.2. Then $\alpha = \Lambda/\Lambda° = 5.2/390.7 = 0.0133$, that is, 1.3%. This is the same number we obtained in the first book from Ka: 1.34%!

Sage: Ostwald's dilution law: $K_a = c\cdot\alpha^2/(1-\alpha) = 0.1\cdot(0.0133)^2/0.9867 = 1.8\cdot10^{-5}$. Conductivity provided an independent method to determine Ka.

Young man: What is the conductivity of pure water? In it, $[H+] = [OH-] = 10^{-7}$ mol/L.

Sage: Calculate: Λ = 349.8 + 198.6 = 548.4 cm·cm²/mol, κ = Λ·c/1000.

Young man: $\kappa = 548.4\cdot10^{-7}/1000 = 5.5\cdot10^{-8}$ S/cm = 0.055 μS/cm. This is the theoretical limit for perfectly pure water; any impurity, even CO_2 from the air, increases conductivity several times.

[UV] Water purity control
The conductivity of ultrapure water at 25 °C is about 0.055 μS/cm (resistance 18.2 MΩ·cm). This is one of the most sensitive methods for monitoring impurities in water.

Diffusion of ions

Sage: Mobility is related to the diffusion coefficient (Nernst–Einstein): $D = \lambda°\cdot RT/(z^2F^2)$, if λ° is in Sm·m²/mol. Let's calculate for Na+.

Young man: $\lambda° = 50.1\cdot10-4$ S·m²/mol. $D = 50.1\cdot10-4\cdot8.314\cdot298/(96\ 485)^2 = 50.1\cdot10-4\cdot2478.6/9.31\cdot10^9 = 1.33\cdot10-9$ m²/s. For H^+ $-9.3\cdot10-9$ m²/s.

Ionic atmosphere and activity

Sage: Why does Λ decrease with the increase of √c? Because each ion is surrounded by a "cloud" of opposite charges — **an ionic atmosphere** (Debye, Hückel, 1923). It slows down the ion and reduces its "chemical efficiency." The thickness of the atmosphere (Debye length) depends on the ionic strength $I = \frac{1}{2}\cdot\Sigma c_i z_i^2$: for a 1:1 electrolyte in water at 25 °C, it is equal to 0.304 nm/√c.

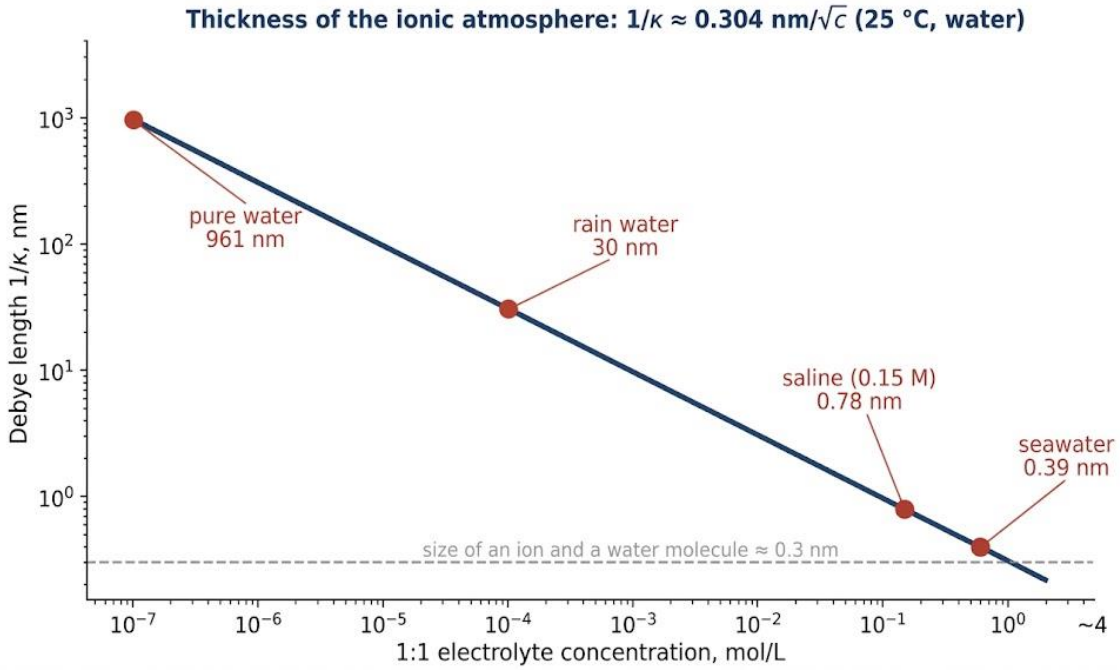


Fig. 11. Thickness of the ionic atmosphere.

Young man: In a saline solution, it is less than a nanometer, while in pure water, it is almost a micrometer. This means that in diluted solutions, ions "feel" each other over huge distances!

Sage: Now the main point for the chemist. Due to interactions, the energy of the ion in solution is lower than in the ideal case, and instead of concentration, the activity a = **γ·c enters** the formulas for equilibria. The limiting law of Debye–Hückel: lg γ± = −0.509·|z+z−|·√I (water, 25 °C). Let's verify this experimentally.

Table 8. Mean activity coefficients: limiting law versus experiment.

Electrolyte	m, mol/kg	γ± by DH / DH	γ± experience / measured
NaCl	0,001	0,964	0,966
NaCl	0,01	0,889	0,902
NaCl	0,1	0,690	0,778
$CaCl_2$	0,001	0,880	0,888
$CaCl_2$	0,01	0,667	0,732

Young man: In diluted solutions, everything adds up. At 0.1 mol/kg, the discrepancy is already 11% — the law stops working.

> **[OD] The Limits of the Extreme Law**
> The Debye-Hückel law is accurate only for ionic strengths below about 0.01 mol/kg. For more concentrated solutions, extended equations (Davis) are used, and for brines and seawater, the Pitzer model is applied. Due to the formation of ion pairs in multivalent electrolytes (for example, MgSO4), the discrepancies are greater.

Sage: Practical consequence: pH measures not the concentration, but the activity of hydrogen ions. For 0.01 M HCl the average γ± ≈ 0.90.

Young man: a(H+) ≈ 0.90·0.01 = 0.0090 mol/L, pH = 2.05, not 2.00. In school calculations, we write 2 because we consider ions to be ideal; in precise measurements, the difference is evident.

> **"Ions in solution behave independently."**
> They are independent only at infinite dilution. Already at 0.01 M, the ionic atmosphere significantly changes conductivity, activity, and equilibrium constants.

Chapter summary

- $\Lambda = 1000\kappa/c$; Kohlrausch's law: $\Lambda = \Lambda^\circ - K\sqrt{c}$; $\Lambda^\circ = \lambda^\circ_+ + \lambda^\circ_-$.
- The proton is passed along (Grotthuss); small ions are hydrated more and move more slowly.
- Ionic atmosphere: $\gamma_\pm < 1$; the Debye–Hückel limiting law works at small I.

Self-check

- Find Λ°(NaCl) and t+(Na+). (126.5; 0.40)
- Estimate $\gamma_\pm$ for 0.005 M $MgSO_4$ ($z_+z_- = 4$). (≈ 0.51; $I = 0.02$)

Glossary

Russian	Английский
electrical conductivity	проводимость
Kohlrausch's law	Kohlrausch’s law
transfer number	транспортное число
ionosphere	ионная атмосфера
ionic strength	ионная сила
activity coefficient	коэффициент активности
Debye length	Дебаевская длина

Chapter 6. Electrode Processes: EMF, Fuel Cells, Batteries

In this chapter

Derivation of the Nernst equation from ΔG · temperature coefficient of EMF · electrolysis voltage · overpotential and the Tafel equation · fuel cells and their losses · why lithium · an honest comparison of methods for hydrogen production, including photocatalysis.

Вывод уравнения Нернста из ΔG · температурного коэффициента ЭДС · напряжения электролиза · переизбытка и уравнения Тафеля · топливные элементы и их потери · почему литий · честное сравнение водородных маршрутов, включая фотокатализм.

From ΔG to volts

Young man: In the first book, we wrote E = E° − (0.0592/n)·lg Q "according to Nernst." Now I want to derive it.

Sage: This is one line. The maximum electrical work of the element is equal to −ΔG, and the work of transferring n moles of electrons through a potential difference E is equal to nFE. Thus, ΔG = **−nFE. Substituting** into ΔG = ΔG° + RT·ln Q (chapter 3): −nFE = −nFE° + RT·ln Q.

Young man: I divide by −nF: **E = E° − (RT/nF)·ln** Q. And at 25 °C RT/F·ln 10 = 0.05916 V — that's where 0.0592 comes from!

Sage: Now the application where entropy is visible. Concentration element: two copper electrodes, one at [Cu2+] = 0.001 M, the other at 0.1 M. There is no chemical reaction, E° = 0.

Young man: Q = 0.001/0.1 = 0.01 (towards dilution), E = (0.05916/2)·lg(100) = 0.0296·2 = 0.059 V. Where does the energy come from if ΔH = 0?

Sage: From the entropy of mixing, as in Chapter 4: the system transitions from different concentrations to the same, and ΔG < 0 is obtained only due to −TΔS. The current here is the "electrical" output of the entropic gain.

Sage: Temperature dependence: (∂E/∂T)p = ΔS/(nF). For H2 + ½O2 → H2O(l) $_{\Delta S =}$ 69.9 − 130.7 − 102.6 = −163.4 J/K.

Young man: (∂E/∂T) = −163.4/(2·96 485) = −0.85 mV/K. At 80 °C E = 1.229 − 0.00085·55 = 1.182 V. Thus, the fuel cell loses voltage when heated.

Sage: And one more important thing: during the ideal operation of the element, heat is inevitably released $_{Q\text{обр}}$ = TΔS = 298·(−163.4) = −48.7 kJ/mol. This is exactly ΔH − ΔG = −285.8 + 237.1. Even an ideal fuel cell does not convert all the heat of the reaction into electricity.

Electrolysis: how many volts are needed

The sage: The reverse reaction requires a minimum Eобр = 1.229 V (ΔG). **But there is also the** thermoneutral voltage Eтн = ΔH/(nF) = 285.8·10^3/(2·96 485) = 1.481 V (higher heat) or 1.253 V (for water vapor, lower heat 241.8 kJ/mol).

[UV] Between 1.23 and 1.48 V
At a voltage between 1.229 and 1.481 V, the electrolyzer, operating reversibly, absorbs heat from the environment (TΔS > 0); above 1.481 V, it releases heat. Practical electrolyzers operate at 1.8–2.0 V.

Young man: I calculate the costs: at 1.9 V per mole H2, 2·96 485·1.9 = 366.7 kJ is needed, that is, 367 kJ/2.016 g = 182 MJ/kg = 50.5 kW·h/kg. And the lower heating value is 33.3 kW·h/kg. Efficiency = 33.3/50.5 = 66 %.

Sage: This is a typical order for modern electrolyzers. Where is the rest?

Overvoltage and the Tafel equation

The sage: Equilibrium thermodynamics states that at Eobr the reaction "can" proceed, but the rate is zero. To generate a current, an overpotential η **must be** applied — above equilibrium. Tafel (1905) empirically found: η = a + **b·lg** j. The theory (Butler–Volmer) explains: b = 2,303RT/(αF), where α ≈ 0.5.

Young man: b = 0.05916/0.5 = 0.118 V per decade of current. Each tenfold increase in current costs 118 mV!

Table 9. Exchange current densities for hydrogen evolution (orders of magnitude, acidic solution).

Electrode	j0 (выделение H2), A/cm² / exchange current density	Comment
Pt	≈ 10^{-3}	best catalyst
Ni	≈ 10^{-5}	good, cheap
Fe	≈ 10^{-5}	corrodes, but active
Cu	≈ 10^{-6}	average / moderate
Pb, Hg	≈ 10^{-12}	very poor catalyst

The young man: Mercury and platinum differ by 109 times! For Hg, to achieve the same current, a higher voltage is needed by 9·0.118 = 1.06 V.

Sage: Therefore, in polarography, the mercury electrode allows operation at very negative potentials: water does not discharge on it. The idea of "good" and "bad" catalysts is key to electrocatalysis, and we will return to it (Part III).

[OD] The values of j0 are approximate.
Exchange currents are strongly dependent on the state of the surface, the solution, and the measurement method; the table provides orders of magnitude to illustrate the variation by billions, rather than reference constants.

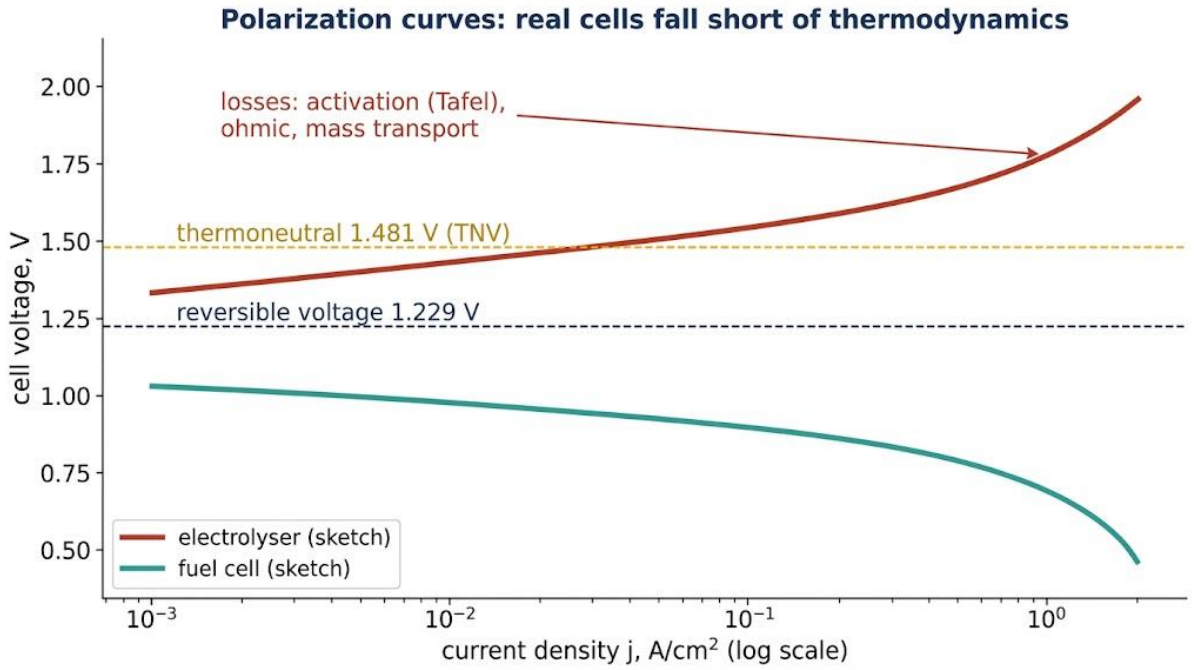


Fig. 12. Polarization curves of an electrolyser and a fuel cell (sketch).

Fuel cells

Sage: In a fuel cell, hydrogen and oxygen react at different electrodes, while electrons flow through an external circuit. The main idea: it is not limited by the Carnot cycle, as chemical energy is converted directly into electrical energy.

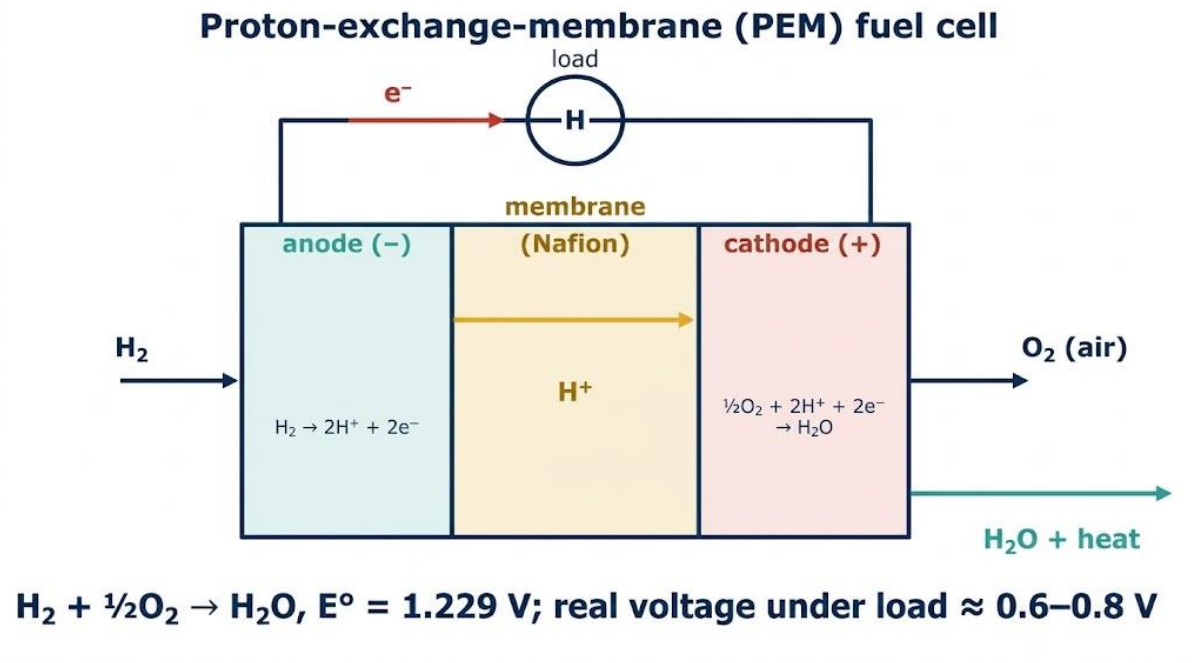


Fig. 13. A proton-exchange-membrane fuel cell.

Table 10. Main fuel-cell types.

Type	Electrolyte	Temperature	Feature
PEMFC	polymer membrane	60–80 °C	fast start, transport
AFC	KOH	60–90 °C	sensitive to CO2
PAFC	H_3PO_4	150–200 °C	stationary plants

Type	Electrolyte	Temperature	Feature
MCFC	molten carbonates	≈ 650 °C	can use CH4 / can use CH4
SOFC	ceramics (ZrO2:Y2O3) / ceramic	600–1000 °C	high efficiency, demanding materials

[OD] Efficiency of the fuel cell
The ideal limit for H2/O2: ΔG/ΔH = 83% (based on higher heat). Real systems operate with an efficiency of about 40–60% due to overvoltages at the electrodes, ohmic losses, and mass transfer (Fig. 12). These figures are approximate and depend on the mode and definition of efficiency.

Batteries: why lithium

Sage: The energy of the battery per unit mass is determined by two things: the voltage E and the charge carried per gram of active material. For a lead-acid battery Pb + PbO2 + 2H2SO4 → 2PbSO4 + 2H2O: E = 2.04 V, n = 2; the mass of the reagents is 207.2 + 239.2 + 196.2 = 642.6 g.

Young man: Charge 2·26.8 Ah = 53.6 Ah. Energy 2.04·53.6 = 109 Wh per 0.643 kg, that is 170 Wh/kg. And real lead batteries provide 30–40.

Sage: Correct, the rest is the mass of the casing, grids, electrolyte, and incomplete utilization. Now let's compare the charge per gram of metal: q = nF/M.

Young man: Li: 96 485/6.94 = 13 900 C/g = 3.86 A·h/g. Zn: 2·26.8/65.4 = 0.82 A·h/g. Pb: 2·26.8/207.2 = 0.26 A·h/g. Lithium is 4.7 times better than zinc and 15 times better than lead, and it has the most negative E° (–3.04 V)!

[UV] Why lithium
Lithium combines the lowest atomic mass among metals and the most negative standard potential: a specific charge of 3.86 A·h/g and a high voltage provide a record specific energy. Modern lithium-ion batteries deliver about 150–250 Wh/kg (depending on the design).

Hydrogen from water: an honest comparison of pathways

Sage: Hydrogen can be obtained from water in three ways: electrolysis (electricity input), thermochemistry (heat), and photocatalysis (light). For each, the term "efficiency" is used, but in all cases, the definition is different, and they cannot be compared directly.

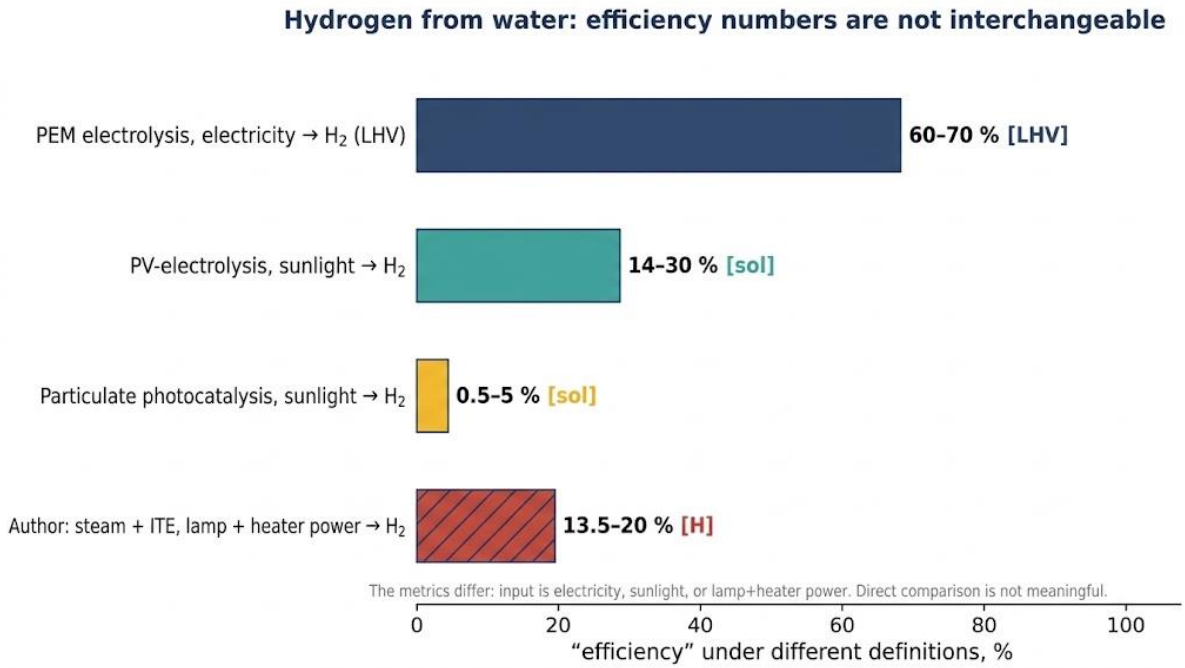


Fig. 14. "Efficiency" of hydrogen production under different definitions.

Sage: Literature benchmarks. Electrolysis with a polymer membrane achieves about 60–70% of the lower heating value of hydrogen (as calculated above). Photoelectrolysis using "solar cell + electrolyzer" yielded 14% and above, with the best demonstrations exceeding 30% for sunlight to hydrogen conversion. Powdered photocatalysts for complete water decomposition typically yield single-digit percentages or less; the apparent quantum efficiency in the best systems reaches up to 95% in the near ultraviolet, but this is the quantum yield at a single wavelength, not the efficiency across the solar spectrum.

The author has their own data on the photocatalytic decomposition of water vapor at 350 and 450 °C: quantum yield ≈ 1 at effective energy 12.6 eV; completeness of vapor processing 85–95%; efficiency 13.5–20% — as the ratio of the energy obtained H2 to the total power consumed during external heating. Let's check if these numbers are consistent with each other and with thermodynamics.

Sage: A good idea: this is how any new results are handled. First, definitions. Quantum **yield Φ** — the number of product molecules per absorbed quantum (or "event"). For the complete decomposition of water, the literature usually states "apparent quantum yield" = 2·N(H2)/N(photons), because two electrons are needed per molecule of H2. Conversion completeness X — **the fraction** of the supplied steam that has reacted. Efficiency **η** — the energy of the produced hydrogen (lower heating value) to the total power consumed.

Sage: Now the energy per molecule H2. Minimum: $\Delta G(H_2O(g) \rightarrow H_2 + \frac{1}{2}O_2) = 228.6$ kJ/mol = 2.37 eV. Lower heat 241.8 kJ/mol = 2.51 eV.

Young man: If one "event" with an effective energy of 12.6 eV produces one molecule H2 (Φ = 1 per molecule), then ηmax = 2.51/12.6 = 19.9%. This is exactly the upper limit of 20% in the author's data! And heating steam to 450 °C: converting 1 mole of water from liquid at 25 °C to steam at 450 °C requires 5.7 + 40.7 + 12.1 = 58.4 kJ, which is 0.61 eV. Then η = 2.51/(12.6 + 0.61) = 19.0%.

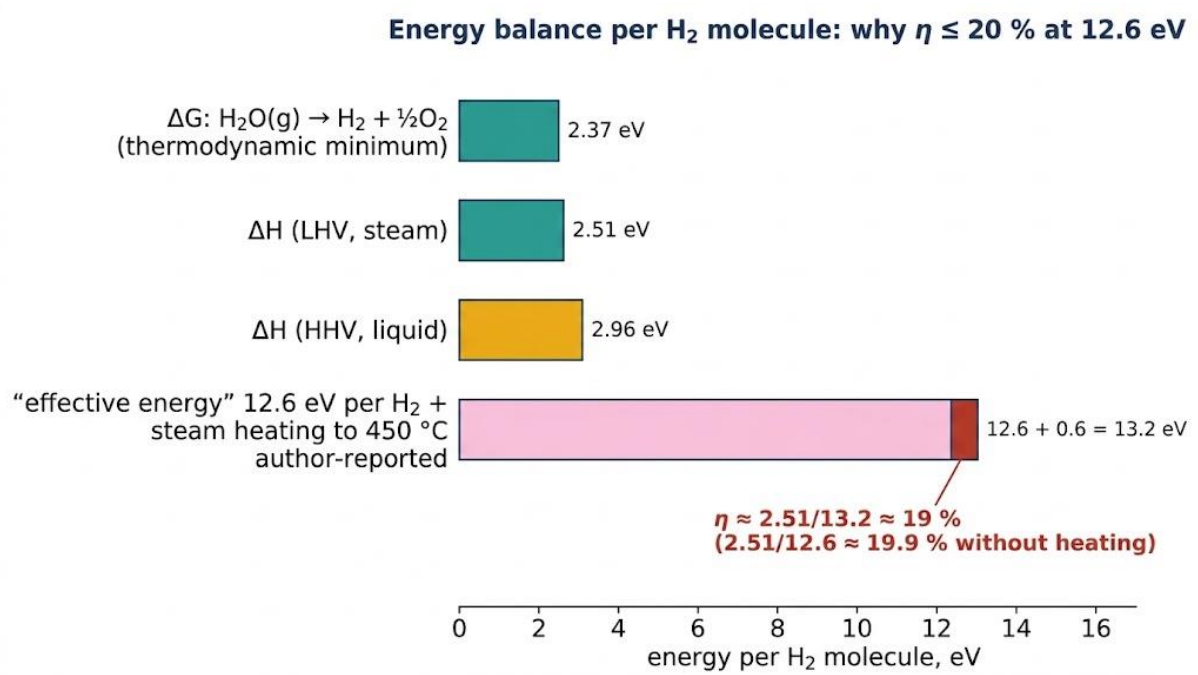


Fig. 15. Energy per H2 molecule: the thermodynamic minimum and the cost of a 12.6 eV event.

Sage: This is important and pleasant. Firstly, the upper limit of 20% is not a random figure, but a direct consequence of how much energy one event "costs." From this, it is clear that **the data do not violate** thermodynamics: η does not exceed the ratio 2.5/12.6. This is consistent with the author's remark that ITE operates through kinetics and overcoming barriers and is limited by the same thermodynamic limit as any method. Secondly, to increase efficiency, it is necessary to reduce the "cost" of the event per molecule H2, rather than waiting for miracles.

Young man: And is the lower limit 13.5%?

The sage: It corresponds to Φ ≈ 0.7 molecules $_{H2}$ per event at the same cost or additional losses (in the lamp, in steam preparation). But here it is necessary to clarify the definitions: what is included in the "total consumed power" — the electrical power of the lamp, optical, or only for heating? And how is "quantum yield ≈ 1" understood: per molecule H2 or according to the formula with the two? If according to the formula with the two, then for one event 12.6 eV corresponds to 0.5 molecules H2, and the upper limit of efficiency would be about 10%.

> **[OD] Definitions solve everything**
> The same measurements yield "quantum outputs" and "efficiencies" that differ by a factor of two depending on what is considered the input and what is considered the output. Therefore, it is necessary to explicitly write the formulas for Φ, X, and η in the publication and specify the method of measuring power.

Sage: Now the completeness is 85–95%. For the thermal decomposition of vapor at 450 °C, the equilibrium degree of decomposition is negligible. Consider: ΔH ≈ 241.8 kJ, ΔS ≈ 44.4 J/K.

Young man: ΔG(723 K) = 241,800 − 723·44.4 = 209.7 kJ. K = exp(−209,700/(8.314·723)) = e−34.9 = $^{7\cdot10-16.}$ For ^{H2O} → H2 + $_{½O2}$ at 1 ^{bar} α1.5 ≈ √2·K, hence $^{\alpha}$ ≈ 10−10. There can be no talk of 85–95% thermally.

Sage: Correct. Thus, high fullness is either a nonequilibrium mechanism (this is precisely what ITE asserts as a hypothesis), or oxygen is leaving the gas phase (for example, oxidizing

the reactor or catalyst material), and then H_2 is obtained, but the reaction is not the decomposition of water, but oxidation. It can be distinguished by a simple balance: in the first case $H_2 : O_2 = 2 : 1$, in the second H_2 is noticeably greater than $2 \cdot O_2$.

[NI] Author data: status
The quantum yield is approximately 1 at 12.6 eV, with completeness of 85–95% (according to the author, hydrogen is confirmed by gas chromatography) and efficiency of 13.5–20% — results from one group. We have not seen independent confirmation, and we cannot consider them an established fact. Internally, they are consistent (efficiency ≤ 20% ≈ 2.51/12.6) and do not contradict thermodynamics.

[Γ] ITE Mechanism
The explanation of high completeness and quantum yield by the impulse tunneling effect (nonequilibrium, "impulse" overcoming of the energy barrier) is the authors' hypothesis. The data do not exclude it, but do not prove it by themselves: the same result could be achieved by other mechanisms.

Table 11. Reported figures and how to check them.

Indicator / Quantity	Author-reported	What to check
quantum yield Φ	≈ 1 at 12.6 eV	formula $\Phi_{(at\ H_2}$ or $_{2N(H_2)/Nquanta)}$; how absorbed energy was measured / formula for Φ; how absorbed energy was measured
completeness X / conversion	85–95 %	balance $H_2 : _{O_2} = 2 : 1$; blank run without catalyst and without impulses / $H_2 : O_2 = 2 : 1$; blank run
η / efficiency	13.5–20 % (350 and 450 °C)	what “total power” includes: lamp, heating; calibrated power meters
upper bound	2,51/12,6 ≈ 19,9 %	matches the top of the range; η does not exceed the limit

Sage: What would make the data resistant to criticism? Control experiment: steam at the same temperature without light and without a catalyst. Isotope labeling: D_2O or $H_2{}^{18}O$ to trace the origins of H_2 and O_2. Oxygen balance. Complete energy balance with calibrated power measurements of the lamp and heater. And most importantly — independent reproduction in another laboratory.

[OD] Our proposals, not criticism
The listed control experiments are a standard set for any report on a new method of hydrogen production. The number 12.6 eV is close to the first ionization energy of a water molecule (12.62 eV); whether this is related to the "effective energy" of the events is a question for the authors, we do not assert any connection.

Chapter summary

- $\Delta G = -nFE \Rightarrow E = E^\circ - (RT/nF)\cdot \ln Q$; $(\partial E/\partial T) = \Delta S/(nF)$. / $\Delta G = -nFE \Rightarrow E = E^\circ - (RT/nF)\cdot \ln Q$; $(\partial E/\partial T) = \Delta S/(nF)$.
- Real cells fall short because of overpotential (Tafel: 118 mV per decade at $\alpha = 0.5$).

- Hydrogen “efficiency” numbers cannot be compared without common definitions; new data need balances and controls.

Self-check

- Find E of Zn|Zn2+(0.01 M)||Cu2+(1 M)|Cu at 25 °C. (1.10 + 0.059 = 1.16 V)
- How many kWh per kg of H2 at 1.7 V? (≈ 45)

Glossary

Russian	Английский
EMF	электромоторная сила (ЭДС)
Nernst equation	Уравнение Нернста
overvoltage	overpotential
Tafel equation	Уравнение Тафеля
exchange current	плотность обменного тока
fuel cell	топливный элемент
quantum output	квантовый выход

Summary of Part II

We saw that voltage is Gibbs energy in different units, that ions in solution interact and "feel" each other, and that real electrochemical systems are defined not only by thermodynamics but also by the rate of electrode reactions. This brings us to the main question: how **fast? This** is the focus of chemical kinetics.

> **What next / Next**
> **Part III. Chemical Kinetics** (Chapters 7–9): reaction rate and order; temperature, Arrhenius equation, and transition state theory; mechanisms, chain reactions, and catalysis.
>
> *Part III: Chemical kinetics (chapters 7–9): rate and order; temperature, the Arrhenius equation and transition-state theory; mechanisms, chain reactions and catalysis.*

PART III. Chemical Kinetics

Часть III. Химическая кинетика

Chapters 7–9

Chapter 7. Reaction Rate: Order and Integrated Laws

In this chapter

Rate and rate law · order of reaction (and why it is not equal to stoichiometry) · integral laws and half-life · how to determine order from data · initial rates method · pseudo-order · diffusion limit and timescale of chemistry.

Rate and rate law · reaction order (and why it is not stoichiometry) · integrated laws and half-life · finding the order from data · the method of initial rates · pseudo-order · the diffusion limit and the time scales of chemistry.

Young man: Sage, in the first two parts we asked "can it go" and "how far will it go." Now — "how fast." Is this a separate science?

Sage: Separate. Thermodynamics does not know time: it compares the beginning and the end. Kinetics studies the path and time. We have already seen: diamond "wants" to become graphite (ΔG < 0), but does not become — the barrier is too high. Now we need the language of speed.

Sage: For the reaction aA + bB → products, **the rate** is defined as v = −(1/a)·d[A]/dt. Empirically, it is most often found that v = k·[A]m·[B]n. This is the rate law; k is **the rate constant,** m and n are the orders with respect to the reactants, and m + n is the overall order.

[OS] "The order is equal to the stoichiometric coefficient"

The order is determined only from experience. For 2N2O5 → 4NO2 + O2, the stoichiometry "indicates" second order, while experience indicates first: v = k[N2O5]. For H2 + Br2 → 2HBr, the order with respect to bromine is fractional (½): v ≈ k[H2][Br2]1/2 at the beginning of the reaction. The reason lies in the mechanism (Chapter 9).

Integral laws

Sage: The rate law is a differential equation. By integrating it, we obtain how the concentration changes over time.

Table 12. The simplest reaction orders.

Order	Rate law	Integrated law	coordinates	t½	Размерность k
0	v = k	[A] = [A]0 − kt	[A] from t / vs t	[A]0/(2k)	M·s−1 / M·s−1
1	v = k[A]	ln[A] = ln[A]0 − kt	ln[A] from t / vs t	ln 2 / k	s−1 / s−1
2	v = k[A]2	1/[A] = 1/[A]0 + kt	1/[A] from t / vs t	1/(k[A]0)	M−1·c−1 / M−1·s−1

Young man: The first-order characteristic is striking: the half-life does not depend on the initial concentration. t½ = ln 2/k.

Sage: Let's test it on a real reaction. The decomposition of N_2O_5 in gas at 45 °C is first order, k = 4.98·10−4 s−1. Find t½ and the fraction remaining after one hour.

Young man: t½ = 0.693/4.98·10−4 = 1392 s ≈ 23 min. After 3600 s: [A]/[A]0 = exp(−4.98·10−4·3600) = exp(−1.79) = 0.17. 17% remains. And at 25 °C k = 3.46·10−5 s^{-1}, $t_{1/2}$ ≈ 5.6 h: with temperature, the rate increases significantly (chapter 8).

How to determine the order

Sage: In practice, the order is unknown. We take the data [A](t) and try three "linearizations": [A] vs. t, ln[A] vs. t, 1/[A] vs. t. The dependence that gives a straight line indicates the order.

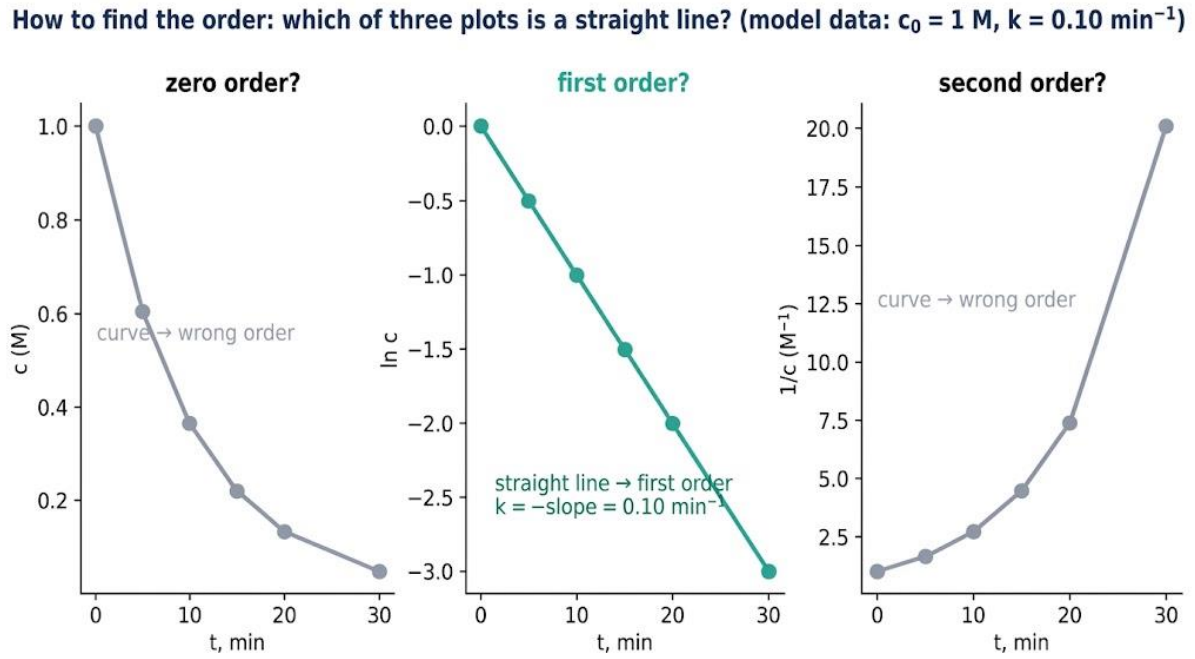


Fig. 16. Finding the order from linearized plots.

Sage: Another method is **the initial rates method:** a series of experiments is conducted with different initial concentrations, and the initial rate is measured.

Table 13. Model data for the method of initial rates.

Run	[A]0, M	[B]0, M	v_0, M/s / M/s
1	0,10	0,10	2,0·10⁻³
2	0,20	0,10	4,0·10⁻³
3	0,10	0,20	8,0·10⁻³

Young man: Experiment 1 → 2: [A] is doubled, v is doubled: first order with respect to A. Experiment 1 → 3: [B] is doubled, v is quadrupled: second order with respect to B. v = $k[A][B]^2$. k = v/([A][B]2) = 2.0·10−3/(0.10·0.010) = 2.0 $M^{-2}·s^{-1}$. The dimensional analysis suggests a total order of 3.

Sage: Another trick: pseudo-order. **If reagent** B is taken in large excess (for example, water during the hydrolysis of an ether), its concentration changes very little, and the law v = k[A][B] behaves like first order: v = k′[A], k′ = k[B].

[UV] Radioactive decay - first order
N = N0e−λt, t½ = ln 2/λ. For ^{14}C, t½ = 5730 years. A sample in which 12.5% of the original 14C remains has lived through three periods: about 17,000 years. This is used in radiocarbon dating.

Time scale

Sage: Reactions occur over a wide range of times. What is the rate of the fastest bimolecular reaction in solution?

Young man: Molecules must first meet. Diffusion limit (Smoluchowski): k_diff = $8RT/(3\eta)$. For water at 298 K η = 8.9·10−4 Pa·s: 8·8.314·298/(3·8.9·10−4) = 7.4·10^6 m³/(mol·s) = 7.4·10^9 M−1·s−1.

The sage: That's right, and this is the ceiling for ordinary reactions: reactions cannot occur faster than molecules meet. The records belong to ions: H+ + OH− → H2O occurs with k = 1.4·10^11 M−1·s−1 — above the limit, because ions are attracted, and the proton "relay races" along hydrogen bonds (chapter 5). These speeds were measured by Eigen using relaxation methods (Nobel Prize 1967).

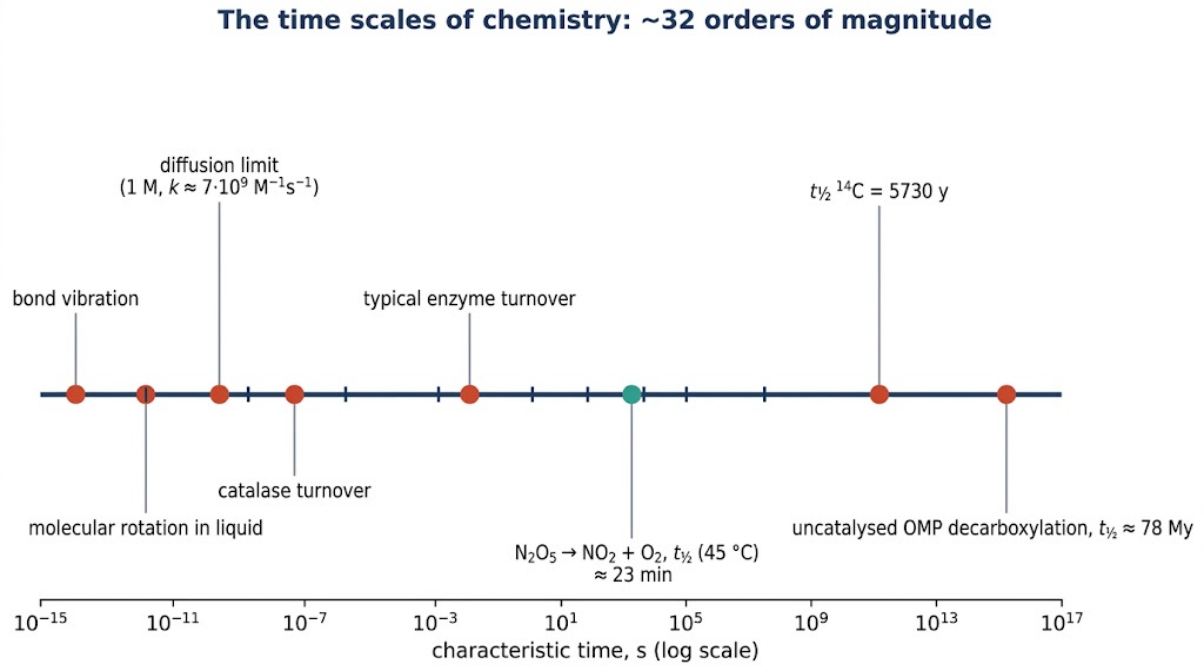


Fig. 17. The time scales of chemistry.

[UV] Femtochemistry
Zewail (Nobel Prize 1999) traced the breaking and formation of bonds "in real time" using laser pulses lasting tens of femtoseconds (10−14 s). This is the lower edge of the scale: the time of one oscillation of a chemical bond.

Chapter summary

- The rate law comes from experiment; order ≠ stoichiometry.

- Zero, first, second orders: ln[A] or 1/[A] vs t are straight lines; first-order t½ is independent of [A]0.
- Initial rates and pseudo-order; the diffusion limit k ≈ 1010 M−1s−1.

Self-check

- t½ = 10 min
- The concentration of a second-order reactant falls from 0.10 to 0.05 M in 100 s. How long to reach 0.025 M? (200 s)

Glossary

Russian	Английский
reaction rate	скорость реакции
rate law	скорость реакции
reaction order	порядок реакции
speed constant	константа скорости
half-life	период полураспада
pseudorder	псевдо-порядок
diffusion limit	диффузионный предел

Chapter 8. Temperature and Theory: Arrhenius, Collisions, the Transition State

In this chapter

Arrhenius equation and activation energy · "double for every 10 degrees" rule · collision theory · transition state theory (Eyring) · relationship between barrier and reaction time · isotopic effect and tunneling.

Уравнение Аррениуса и энергия активации · правило «удвоения на каждые 10 К» · теория столкновений · теория переходного состояния (Эйринг) · барьер и время реакции · изотопные эффекты и туннелирование.

Young man: Sage, a pressure cooker cooks faster. But why does a difference of only 20 degrees provide such an advantage?

Sage: Because the rate depends on temperature exponentially. Arrhenius (1889) found: k = A·exp(−Ea/RT). Ea is **the activation energy, A** is the pre-exponential factor. If you plot ln k against 1/T, you will get a straight line with a slope of −Ea/R.

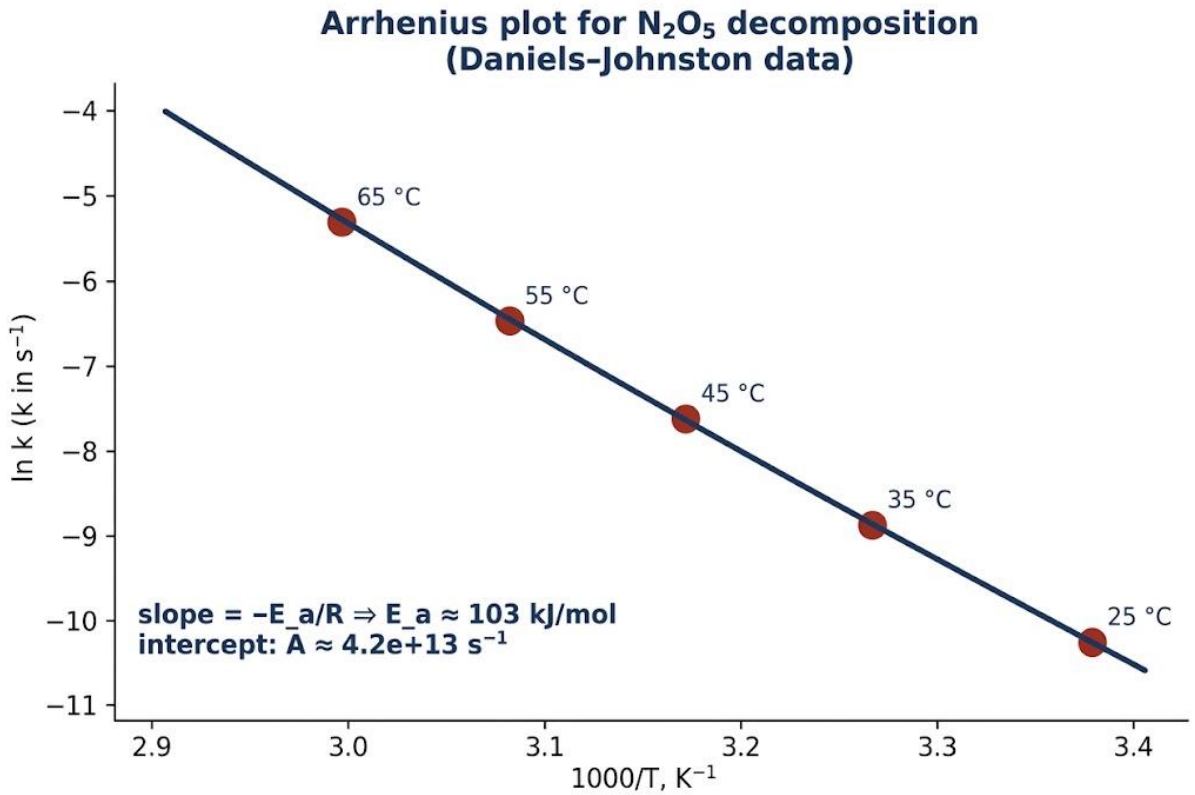


Fig. 18. Arrhenius plot for N2O5 decomposition.

Young man: For N2O5, from the slope, we get Ea ≈ 103 kJ/mol, and A ≈ 4·10^13 s−1. This is the activation energy of the reaction, and it is 100 times greater than the average thermal energy RT = 2.5 kJ/mol.

Sage: Here is what it gives. By how many times will the speed increase when heated from 298 to 308 K?

Table 14. Rate increase for +10 K (from 298 K).

Ea, kJ/mol	k(308 K)/k(298 K)
25	1,4
50	1,9
75	2,7
100	3,7

Young man: So, "twice at 10 degrees" is the rule for Ea around 50 kJ/mol. And in the pressure cooker: from 100 to 121 °C with Ea = 80 kJ/mol: ln(k2/k1) = (80 000/8,314)·(1/373,15 − 1/394,15) = 9622·1,43·10−4 = 1,37, that is, 3.9 times. Almost 4 times faster — as it should be in a pressure cooker.

Collision theory: where the exponent comes from

Sage: For molecules to react, they need to meet and possess energy of at least Ea. According to Boltzmann, the fraction of molecules with energy above Ea is proportional to exp(−Ea/RT).

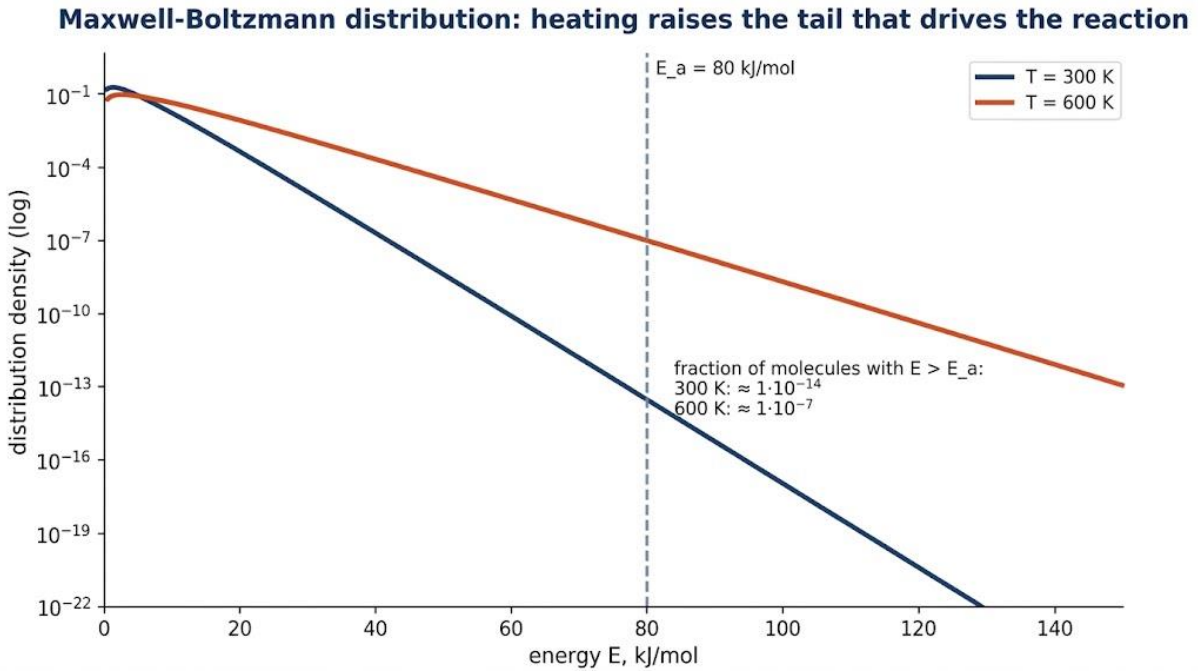


Fig. 19. The high-energy tail and its growth on heating.

Young man: For Ea = 80 kJ/mol at 300 K, the fraction e−32 ≈ 10−14; at 600 K — e−16 ≈ 10−7. A billion-fold increase with a twofold increase in temperature! And there are many collisions: about 10^9 per second per gas molecule under normal conditions.

Sage: Correct. The collision theory gives k = P·Z·exp(−Ea/RT), where Z is the collision frequency, P is the steric factor (not every encounter has a successful orientation). It works well for simple gas-phase reactions, but it does not explain why P can be 10−6, and it does not utilize knowledge about the structure of molecules. This was accomplished by the transition state theory.

Transition state theory (Eyring, 1935)

Sage: The idea: there is a "pass" on the energy surface between the reactants and **products — the activated** complex, the transition state (‡). It is in quasi-equilibrium with the reactants and decays with a frequency of kBT/h. Then **k = (kBT/h)·exp(−ΔG‡/RT), where ΔG‡ =** ΔH‡ − TΔS‡.

Young man: kBT/h = 1.381·10−23·298/6.626·10−34 = 6.2·10^12 s−1. This is a frequency of oscillations on the order of thermal! And A for N2O5 is about 4·10^13: the same order. Is this coincidence not accidental?

Sage: No. For a monomolecular reaction A = e·(kBT/h)·exp(ΔS‡/R). From A = 4.2·10^13 we get exp(ΔS‡/R) = 4.2·10^13/(2.718·6.2·10^12) = 2.5, from which ΔS‡ ≈ +8 J/(mol·K): the transition state is slightly "looser" than the reactant. And ΔH‡ = Ea − RT = 103 − 2.5 ≈ 101 kJ/mol.

> **[OD] The sign ΔS‡ as information about the mechanism**
> For reactions involving the addition of two molecules into a single transition state, ΔS‡ is strongly negative (with a pre-exponential factor of 10^8–10^10 M−1s−1), while for dissociation it is around zero or positive. From A, one can infer how tightly ordered the transition state is.

Sage: Now the main thing: the relationship between the barrier and time. Take four values of ΔG‡ at 298 K.

Table 15. Barrier and lifetime at 298 K (from the Eyring equation).

ΔG‡, kJ/mol	k, s⁻¹ / s⁻¹	t½
60	190	3.6 ms / ms
80	0,058	12 s / s
100	1,9·10⁻⁵	10 h / h
120	6,1·10⁻⁹	3.6 years

Young man: Every 20 kJ/mol — three to four orders of magnitude in time. And every 5.7 kJ/mol (RT·ln 10) — exactly ten times!

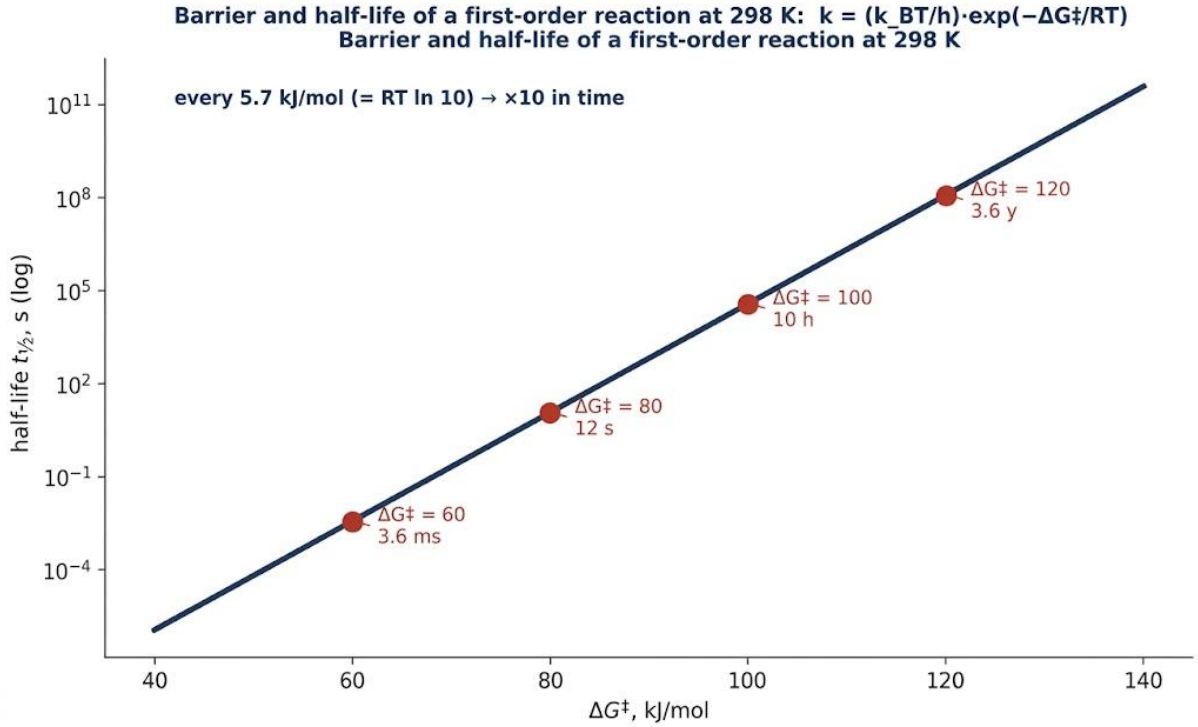


Fig. 20. The barrier ΔG‡ and half-life at 298 K.

[UV] What does this scale show?
Reactions with a barrier of about 60 kJ/mol finish in milliseconds, around 100 in hours, and about 120 in years. Therefore, diamond and sugar are stable at room temperature, although thermodynamics favors their oxidation.

Isotopic effect and tunneling

Sage: The transition state theory provides a testable prediction: the substitution of H with D changes the rate because the zero-point energy of the C–D bond vibration is lower than that of C–H. Estimate: ν(C–H) ≈ 2900 $^{cm-1,}$ ν(C–D) ≈ 2100 $^{cm-1.}$

Young man: The difference in zero energies ½·hc·Δν = $^{½·800}$ cm−1·0.01196 kJ/mol per cm−1 = 4.8 kJ/mol. The ratio of speeds $_{kH/kD}$ = exp(4.8/2.479) = e1.93 $^{=}$ 6.9. Seven times!

Sage: This is the "classical" maximum of about 7 at 298 K. If the experiment yields much more (enzymes can have dozens), it means the particle overcomes the **barrier through tunneling** — a quantum effect that we will discuss in Part IV.

[UV] Tunneling of light particles
Quantum tunneling of hydrogen and electrons has been experimentally established: it manifests in abnormally large isotopic effects and in the weak dependence of the rate on temperature at low T. Isotopic effects on the order of tens are observed for proton transfer in enzymes (for example, in lipoxygenase).

[Г] Tunneling and the ITE Hypothesis
The author's hypothesis of ITÉ (impulse tunneling effect) suggests that under impulse action on functional ceramics, tunneling transitions across energy barriers may contribute significantly differently from thermal contributions. Within the framework of transition state theory, this would manifest as a deviation from the Eyring equation: a weak "Arrhenius" dependence on T and an unusually large isotopic effect H/D. These are testable predictions; until they are tested, we classify the statement as a hypothesis.

Chapter summary

- $k = A \cdot \exp(-Ea/RT)$; ln k vs 1/T is a straight line; +10 K gives ×2 at Ea ≈ 50 kJ/mol.
- $k = (kBT/h) \cdot \exp(-\Delta G^{\ddagger}/RT)$: the lifetime depends exponentially on the barrier; 5.7 kJ/mol → ×10.
- The H/D isotope effect ≈ 7 is the classical limit; larger values indicate tunneling.

Self-check

- Find Ea if the rate triples on heating from 300 to 310 K. (≈ 85 kJ/mol)
- Estimate t½ for $\Delta G^{\ddagger}$ = 90 kJ/mol at 298 K. (≈ 11 min)

Glossary

Russian	Английский
activation energy	активационная энергия
pre-exponential factor	pre-exponential factor
activated complex	активированный комплекс
transitional state	переходное состояние
isotopic effect	изотопный эффект
tunneling	tunnelling

Chapter 9. Mechanisms and Catalysis

In this chapter

Elementary stages and molecularity · rate-determining step · method of steady-state concentrations · Lind mechanism · chain reactions · catalysis: what it changes and what it does not · enzymes and the Michaelis–Menten equation · heterogeneous catalysis and the Sabathé "volcano."

Elementary steps and molecularity · the rate-determining step · the steady-state approximation · the Lindemann mechanism · chain reactions · catalysis: what it changes and what it does not · enzymes and the Michaelis–Menten equation · heterogeneous catalysis and Sabatier's volcano.

Young man: Sage, the law of speed v = k[N2O5] is just empiricism. I want to know what happens with the molecules.

Sage: This is **the** mechanism: a sequence of **elementary stages** — reactions occurring "in one step," without intermediate particles. For an elementary stage, the order equals **the molecularity** (the number of colliding particles): unimolecular — first order, bimolecular — second. For a complex reaction, this correspondence does not exist, and the rate law only restricts the possible mechanisms.

Sage: Often one stage — the slowest one — sets the **overall rate: the rate-determining step.** But be cautious: when there are many intermediate particles, it is necessary to use the steady-state approximation instead: to assume that the concentration of the active intermediate particle I quickly reaches a constant level, d[I]/dt ≈ 0.

Monomolecular reactions: Lind.

Young man: Paradox: the monomolecular reaction A → P must be first order. But where does the molecule get its energy? From collisions — and that is a bimolecular process!

The sage: Lindemann (1922) proposed: A + M ⇌ A* + M (the collision produces an excited molecule), A* → P. We apply the steady-state approximation to A*: [A*] = k1[A][M]/(k−1[M] + k2). The rate v = k2[A*].

Young man: At high pressure (k−1[M] ≫ k2) v = (k1k2/k−1)[A] — first order. At low pressure (k−1[M] ≪ k2) v = k1[A][M] — second order, and the rate is proportional to the pressure. The order depends on the pressure!

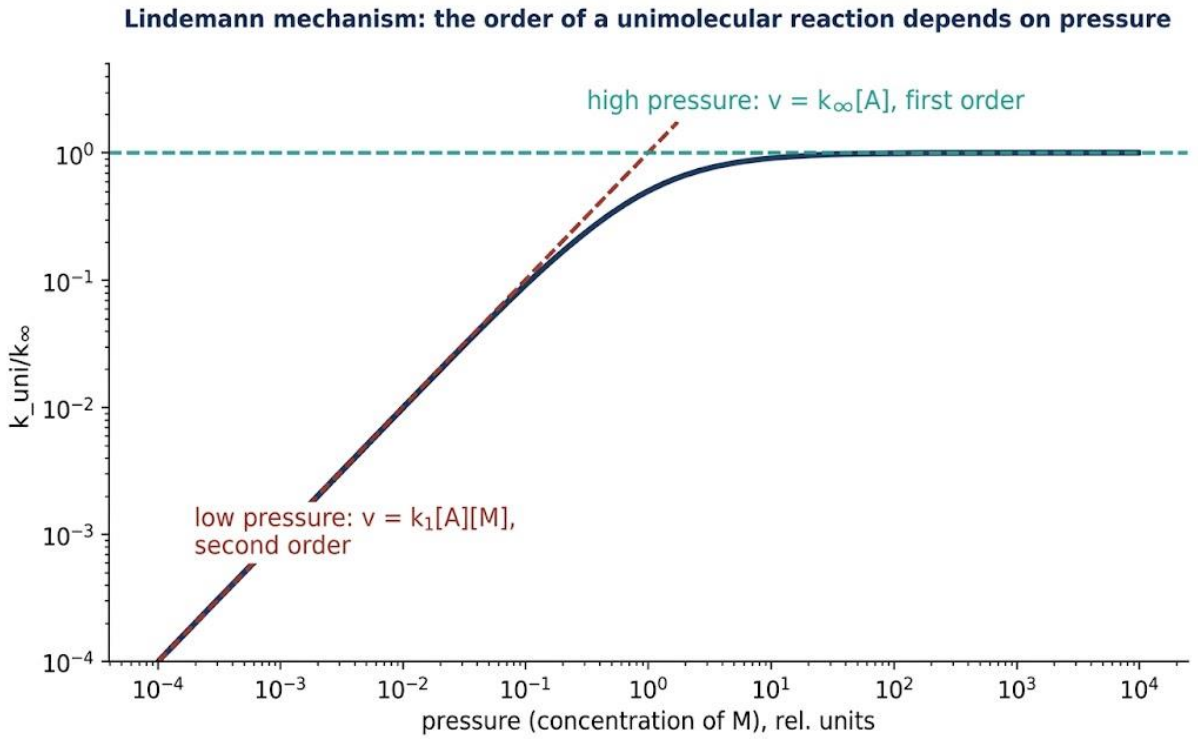


Fig. 21. The Lindemann mechanism: from second to first order.

[UV] The fall of k with pressure
The decrease of the effective constant of a unimolecular reaction with decreasing pressure ("fall-off") is observed experimentally; modern theories (RRKM, Marcus) refine the Lind mechanism but retain its essence.

Chain reactions

The sage: For H2 + Br2 → 2HBr, Bodenstein (1906) found $v = \frac{k[H_2][Br_2]^{1/2}}{(1 + k'[HBr]/[Br_2])}$. He explains this with a chain mechanism: $Br_2 \rightarrow {}_{2Br\cdot}$ (initiation), Br· + H2 → HBr + H·, $H\cdot + {}_{Br_2} \rightarrow$ HBr + Br· (two chain propagation steps), H· + HBr → H2 + Br· (inhibition), 2Br· → Br2 (termination).

Young man: Fractional order ½ with respect to bromine — from equilibrium Br2 ⇌ 2Br·: $[Br\cdot] \propto [Br_2]^{1/2}$. And the inhibition by the product explains the denominator.

Sage: If there is a branching **in the** chain (one particle generates two), the number of active particles grows in an avalanche-like manner — this is an explosion. This is how hydrogen burns in oxygen: $_{H\cdot} + O_2 \rightarrow$ OH· + O·, O· + H2 → OH· + H·. Semenov (Nobel Prize 1956, together with Hinshelwood) created the theory of chain reactions and explained the limits of ignition: at low pressure, the chains break on the walls, while at high pressure — in the volume.

[OD] Safety
Mixtures of hydrogen with oxygen and air are explosive over a wide range of compositions (in air approximately from 4 to 75 % H2). Experiments with them should only be conducted in a specially equipped laboratory.

Catalysis: what it changes and what it does not

The sage: **A catalyst** accelerates a reaction without being consumed: it opens another pathway with a lower $\Delta G^{\ddagger}$. However, it does not shift the equilibrium: K = kпр/kобр, and the catalyst accelerates both directions by the same factor. If you see "the catalyst increased the yield" — it means it accelerated the yield to equilibrium or changed the selectivity, but not K.

> **[UV] Catalyst and Thermodynamics**
> The catalyst does not change ΔG of the reaction and K. It lowers the barrier for both directions. Therefore, the catalyst for ammonia synthesis (Fe) equally accelerates the decomposition NH3.

Sage: What gain? Calculate for a barrier reduction of 20 kJ/mol at 298 K.

Young man: exp(20 000/2478) = e8.07 ≈ 3200 times. And a decrease of 100 kJ/mol — by $10^{17.5}$ times!

The sage: This is exactly how much the enzyme orotidine-5′-monophosphate decarboxylase provides: without it, the reaction has a t½ of about 78 million years, with it — fractions of a second (an acceleration of about 10^{17}). Enzymes are the champions of catalysis.

Table 16. Turnover numbers of some enzymes (approximate).

Enzyme	kcat, s^{-1}	Comment
catalase	≈ $4\cdot10^{7}$	decomposes H2O2; near diffusion limit
carbonic anhydrase	≈ 10^{6}	$CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+$
acetylcholinesterase	≈ $1{,}4\cdot10^{4}$	nervous system
OMP decarboxylase	≈ 40	acceleration ~1017 / rate enhancement ~1017
lysozyme	≈ 0,5	slow but sufficient

The quantitative work of the enzyme is described by the Michaelis–Menten equation: $E + S \rightleftharpoons ES \rightarrow E + P$. In the steady-state approximation, $v = V_{max}\cdot[S]/(K_m + [S])$, $V_{max} = k_{cat}\cdot[E]_0$.

Young man: At [S] = Km, the rate is half of the maximum. And at [S] = 10·Km: 10/11 = 91% Vmax. Further addition of substrate is pointless — the enzyme is "occupied."

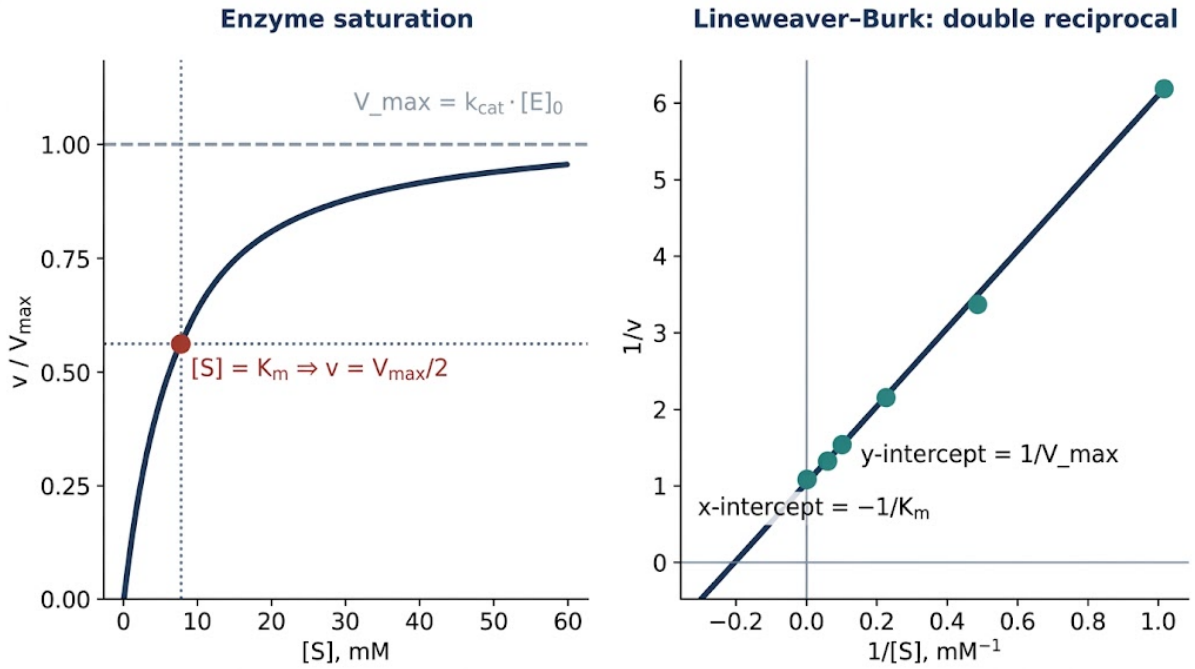


Fig. 22. The Michaelis–Menten equation and the Lineweaver–Burk plot.

Heterogeneous catalysis and the Sabatier principle

Sage: In industry, solid catalysts are more commonly used: Fe in ammonia synthesis (chapter 8 of the first book), Pt–Rh in the Ostwald process, V_2O_5 in the contact method. The reaction occurs in three steps: adsorption of reactants, reaction on the surface, desorption of products. Sabatie formulated the principle: the bond with the surface should **be neither too weak nor too strong.**

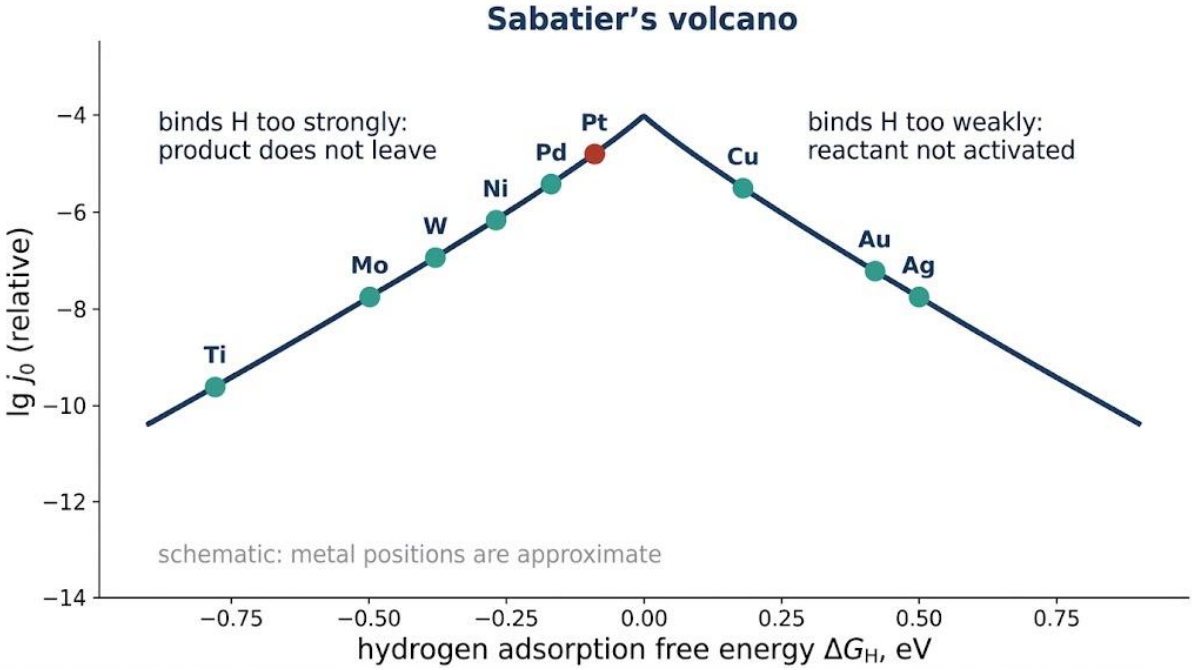


Fig. 23. Sabatier's volcano: activity versus adsorption energy.

Young man: Therefore, in Table 9 (Chapter 6), platinum is the best catalyst for hydrogen evolution, while mercury is the worst: it hardly binds hydrogen. And too strong a bond "poisons" the catalyst — like CO on platinum.

[OD] Volcano diagram — qualitative
The positions of metals on the "volcano" are approximate and depend on calculations and conditions. The Sabatie principle qualitatively explains the optimum; modern theory (Nørskov et al.) refines it using adsorption energy calculations (Part IV).

Chapter summary

- A mechanism is a sequence of elementary steps; the rate law only constrains possible mechanisms.
- The steady-state approximation yields rate laws (Lindemann, chain reactions).
- A catalyst lowers the barrier but does not change K; enzymes and solid catalysts obey saturation and Sabatier's principle.

Self-check

- At what [S] does an enzyme run at 80 % $_{of}$ Vmax if Km $_{=}$ 2 mM? (8 $_{mM}$ / $_{mM)}$
- A catalyst cannot shift the equilibrium because it accelerates both the forward and reverse reactions equally, without changing the position of the equilibrium. It lowers the activation energy for both reactions, allowing them to reach equilibrium faster, but does not alter the concentrations of the reactants and products at equilibrium.

Glossary

Russian	Английский
elementary stage	элементарный шаг
molecularity	молекулярность
stationary approximation	steady-state approximation
chain reaction	цепная реакция
catalyst	катализатор
reciprocal number	число оборота
Michaelis constant	Michaelis constant

Summary of Part III

Kinetics added time to thermodynamics. We saw that the rate is determined by the barrier ΔG‡ exponentially, that temperature "raises the tail" of the distribution, that the mechanism is hidden behind the rate law, and that the catalyst changes the pathway but not the endpoint. The main question remains: **why** are the barriers and connections exactly like this? Quantum chemistry answers this question.

What next / Next

Part IV. Quantum Chemistry (Chapters 10–12): molecular orbitals and the Hartree–Fock method; potential energy surfaces and reaction calculations; spectra and photochemistry. Then — the final conclusion of the book.

Part IV: Quantum chemistry (chapters 10–12): molecular orbitals and the Hartree–Fock method; potential energy surfaces and computing reactions; spectra and photochemistry. Then the book's final conclusion.

PART IV. Quantum Chemistry

Часть IV. Квантовая химия

Chapters 10–12 · final conclusion

Chapter 10. From the Schrödinger Equation to the Molecule

In this chapter

Born-Oppenheimer approximation and potential energy curve · variational principle and MO-LCAO method · π-systems according to Hückel and delocalization energy of benzene · Hartree-Fock method and correlation energy · ladder of methods: HF, DFT, MP2, CCSD(T).

Приближение Борна–Оппенгеймера и кривая потенциальной энергии · вариационный принцип и LCAO–MO · системы Хюккеля π и энергия делокализации бензола · метод Хартри–Фока и энергия корреляции · лестница методов: HF, DFT, MP2, CCSD(T).

Young man: Sage, in the first book, the hydrogen atom was solved "head-on." Why can't the Schrödinger equation for the water molecule be solved in the same way?

Sage: Because for a system of three or more interacting particles, there is no exact solution. For H_2^+ (one electron, two nuclei), the equation is solved exactly, but for H2 (two electrons), we have to approximate. The first approximation is the most important: **Born–Oppenheimer (1927). The** nucleus is 1836 times heavier than the electron, so the electrons "adjust" to the position of the nuclei instantaneously. We solve the electronic problem with **fixed nuclei and** obtain the energy E(R) as a function of distance.

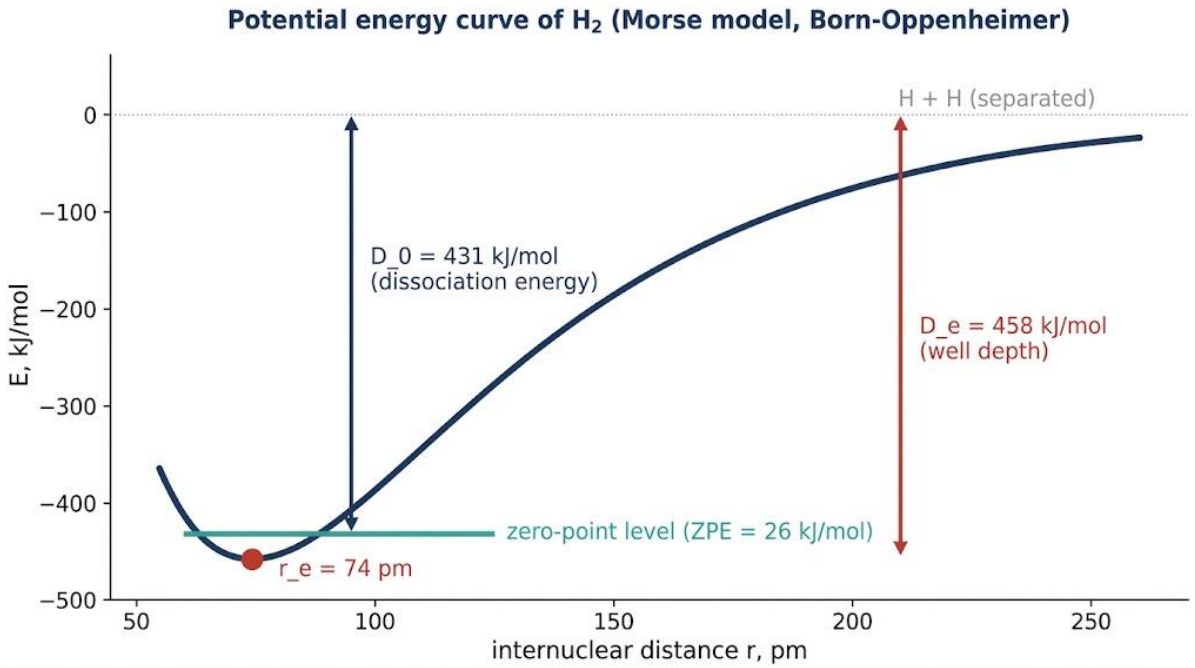


Fig. 24. The potential energy curve of H2.

Young man: So, the "bond length" of 74 pm is the minimum of this curve! And the depth of the well De = 458 kJ/mol. But in the bond table from the first book, H–H was 436.

The sage: The difference is explainable. The molecule does not lie at the bottom of the well: it has vibrations, and even at 0 K, there remains zero energy $\frac{1}{2}h\nu$. $\nu\square$(H2) = 4401 cm−1. Calculate.

Young man: ZPE = ½·4401·0,01196 kJ/mol per cm−1 = 26.3 kJ/mol. Then D0 = 457.6 − 26.3 = 431.3 kJ/mol — the bond dissociation energy from the zero level. Let's add thermal corrections at 298 K — and we get the tabulated value of 436.

> **[UV] Structure of a molecule from quantum mechanics**
> The concepts of "bond length," "valence angle," and "force constant" are properties of the potential energy surface E(R) that arise in the Born-Oppenheimer approximation. It works well for ground states but breaks down near intersections of electronic surfaces (conical intersections), which are important in photochemistry (Chapter 12).

MO–LKAO and the variational principle

Sage: How to find the wave function of electrons? We take a trial function as a linear combination of atomic orbitals (LCAO): ψ = cAφA + **cBφB. Variational principle: the** energy of any trial function is not lower than the true energy of the ground state, so we adjust the coefficients by minimizing E. For two identical orbitals, we obtain E± = (α ± β)/(1 ± S): the bonding orbital is lower, the antibonding is higher. Here, α is the Coulomb integral, β is the resonance integral (negative), S is the overlap.

Sage: For conjugated systems, there is a simple model — the Hückel **method (1931):** only π-electrons are considered and E = α + xβ is calculated, where x are the eigenvalues of the connectivity matrix. For a chain of n atoms, xj = 2cos(jπ/(n + 1)); for benzene, x = 2, 1, 1, −1, −1, −2.

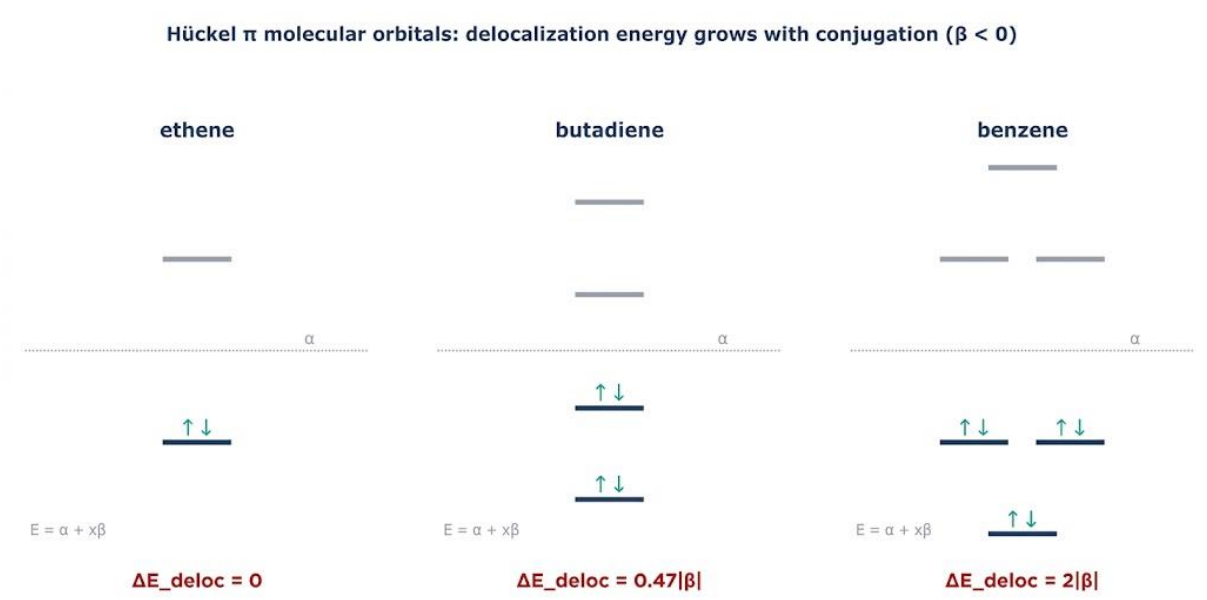


Fig. 25. Hückel π orbitals.

Young man: I calculate the π-energy of benzene: 2(α + 2β) + 4(α + β) = 6α + 8β. And three isolated double bonds of ethylene: 3·2(α + β) = 6α + 6β. The gain is 2β, that is, 2|β|: this is the energy of delocalization.

Sage: Now let's compare with experience. The heat of hydrogenation of cyclohexene is −120 kJ/mol, while that of benzene to cyclohexane is −208.

Young man: Three double bonds "without resonance" would give −360. In reality, it is −208. The difference is 152 kJ/mol. Thus, $|\beta| \approx 76$ kJ/mol. And for butadiene, Hückel gives $0.47|\beta| \approx 36$ kJ/mol.

The sage: The experience: the heat of hydrogenation of butadiene gives about 13–15 kJ/mol of stabilization. The model overestimates by two to three times. This is normal: it is qualitative and does not claim to be quantitative. The parameter β is selected anew for each property: for spectra, it is quite different (on the order of 2.5–3 eV).

> **[OD] What does the Hückel model provide?**
> Correct trends: the longer the conjugated chain, the narrower the gap between the upper occupied and lower unoccupied orbitals, and the longer the absorption wavelength. Ethene absorbs around 165 nm, butadiene — 217 nm, hexatriene — 258 nm, β-carotene — about 450 nm (the orange color of carrots). Quantitatively, the model overestimates the red shift.

Hartree–Fock method and correlation energy

Sage: For a multi-electron molecule, the mean field idea **(Hartree, Fock)** is used: each electron moves in the averaged field of the nuclei and other electrons. The wave function is a Slater determinant, which automatically takes into account the Pauli principle. The equations are solved iteratively until self-consistency (SCF) is achieved.

Young man: But the "mean field" does not take into account that electrons instantly avoid each other?

Sage: Exactly. The difference between the exact energy and the Hartree–Fock energy is called the **correlation energy. For the** helium atom: $E_{HF} = -2.8617$ hartree, the exact value is −2.9037 hartree. One hartree is equal to 27.211 eV = 2625.5 kJ/mol.

Young man: The difference 0.0420 Hartree · 2625.5 = 110 kJ/mol. Hartree-Fock gives 99% of the total energy, but the missing percentage is 110 kJ/mol, which is more than the typical activation energy!

Table 17. Main electronic-structure methods (approximate).

Method	Rise in cost / Cost	Includes	Typical error ΔE of the reaction
Hartree–Fock / HF	$\propto N^4$	mean field, exchange	> 10 kcal/mol / kcal/mol
DFT (hybrid)	$\propto N^3$–N^4	correlation via density functional	2–5 kcal/mol / kcal/mol
MP2	$\propto N^5$	perturbation theory	3–5 kcal/mol / kcal/mol
CCSD(T)	$\propto N^7$	“gold standard”: coupled cluster	≈ 1 kcal/mol / kcal/mol

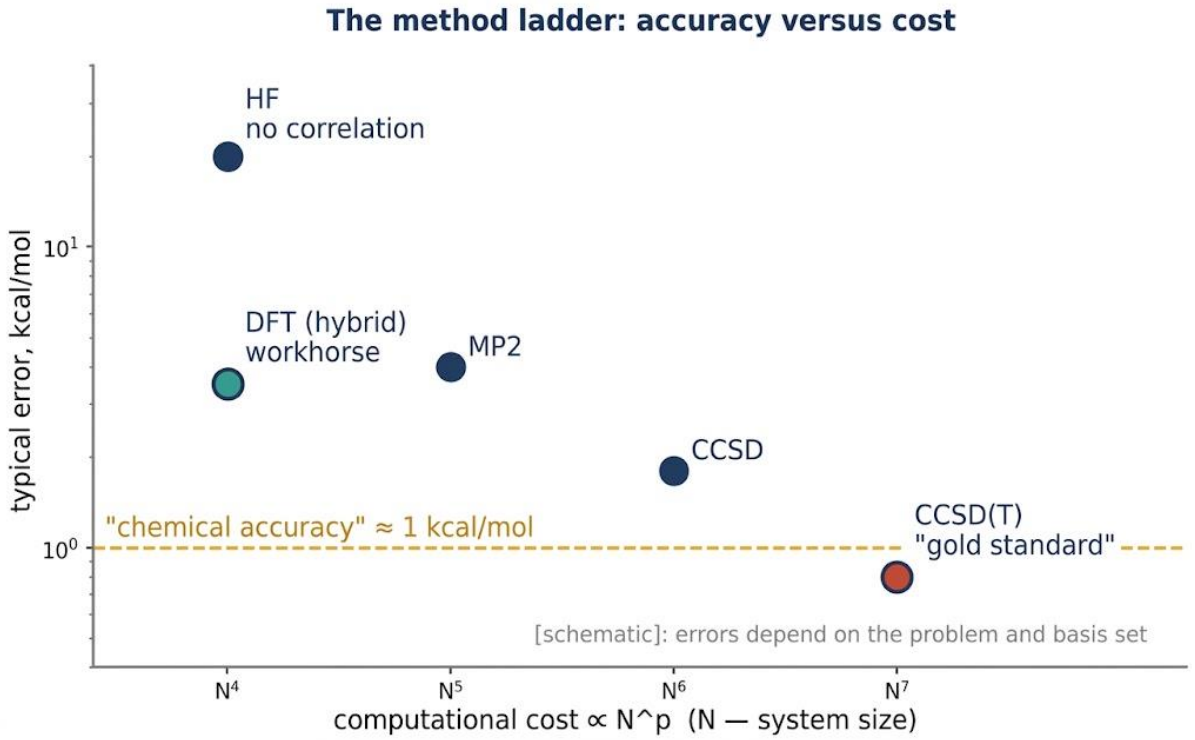


Fig. 26. Accuracy and cost of the methods.

The sage: Density functional theory (DFT) is our working tool: Hohenberg and Kohn (1964) proved that the ground state energy is determined by the electron density, Kohn and Sham (1965) proposed a practical scheme. Kohn and Pople received the Nobel Prize in 1998. All "functionals" in DFT are approximations, and the accuracy depends on the choice.

> **[OD] Error figures — benchmarks**
> Typical errors in Table 17 depend on the class of problems, the basis, and the functionality; these are orders of magnitude, not guarantees. Always compare with a known experimental value for a similar system.

Table 18. Energy units in quantum chemistry.

Unit	eV / eV	kJ/mol	kcal/mol	cm−1 / cm−1
1 hartree / 1 hartree	27,2114	2625,50	627,51	219 474,6
1 eV / 1 eV	1	96,485	23,061	8065,5
1 kJ/mol	0,010364	1	0,2390	83,59
1 kcal/mol / 1 kcal/mol	0,043364	4,184	1	349,76
1 $^{cm-1}$ / 1 $^{cm-1}$	$1{,}2398 \cdot 10^{-4}$	0,011963	0,0028591	1

Chapter summary

- The Born–Oppenheimer approximation gives E(R): the bond $_{length}$ is the minimum, $D_0 = D_e - ZPE$.
- LCAO–MO and Hückel explain delocalization and colour; Hartree–Fock leaves the correlation energy (110 kJ/mol for He).
- The ladder HF → DFT → MP2 → CCSD(T): accuracy grows with cost.

Self-check

- Estimate the ZPE for D2, if ν□(H2) = 4401 cm−1 (ν∝ μ−1/2). / Estimate the ZPE of D2. (≈ 18.6 kJ/mol)
- Find the π energy and the delocalization energy of the allyl cation (x = 1.414; 0; −1.414), 2 electrons. / Find the π energy of the allyl cation. (2α + 2.83β; DE = 0.83|β|)

Glossary

Russian	Английский
Born-Oppenheimer approximation	Born–Oppenheimer approximation
variational principle	вариационный принцип
Hartree–Fock method	Hartree–Fock метод
correlation energy	корреляционная энергия
density functional theory	теория функционала плотности
zero-point energy	ноль-точечная энергия
delocalization energy	делокализационная энергия

Chapter 11. The Potential Energy Surface and Computing Reactions

In this chapter

Potential energy surface (PES) · minima and saddle points · imaginary frequency and tunneling · how to obtain ΔG‡ and k from calculations · error margin: 5.7 kJ/mol — ×10 · catalysis, enzymes, and QM/MM.

Потенциальная энергия поверхности (PES) · минимумы и седловые точки · мнимая частота и туннелирование · от расчета до ΔG‡ и k · цена ошибки: 5.7 кДж/моль → ×10 · катализация, ферменты и QM/MM.

Young man: Sage, in chapter 8 we introduced the "transitional state" as a pass. Where is it actually located?

The sage: **On the surface of potential energy** — the multidimensional "relief" E(q1, q2, ...), where q represents the coordinates of the nuclei (for a molecule with N atoms — 3N – 6 coordinates). Molecules and reactions **are** the geography of this surface. Minima are stable substances (reactants, products, **intermediates). A first-order saddle point** is a pass: a minimum in all directions except one (along which it is a maximum). This is the transition state.

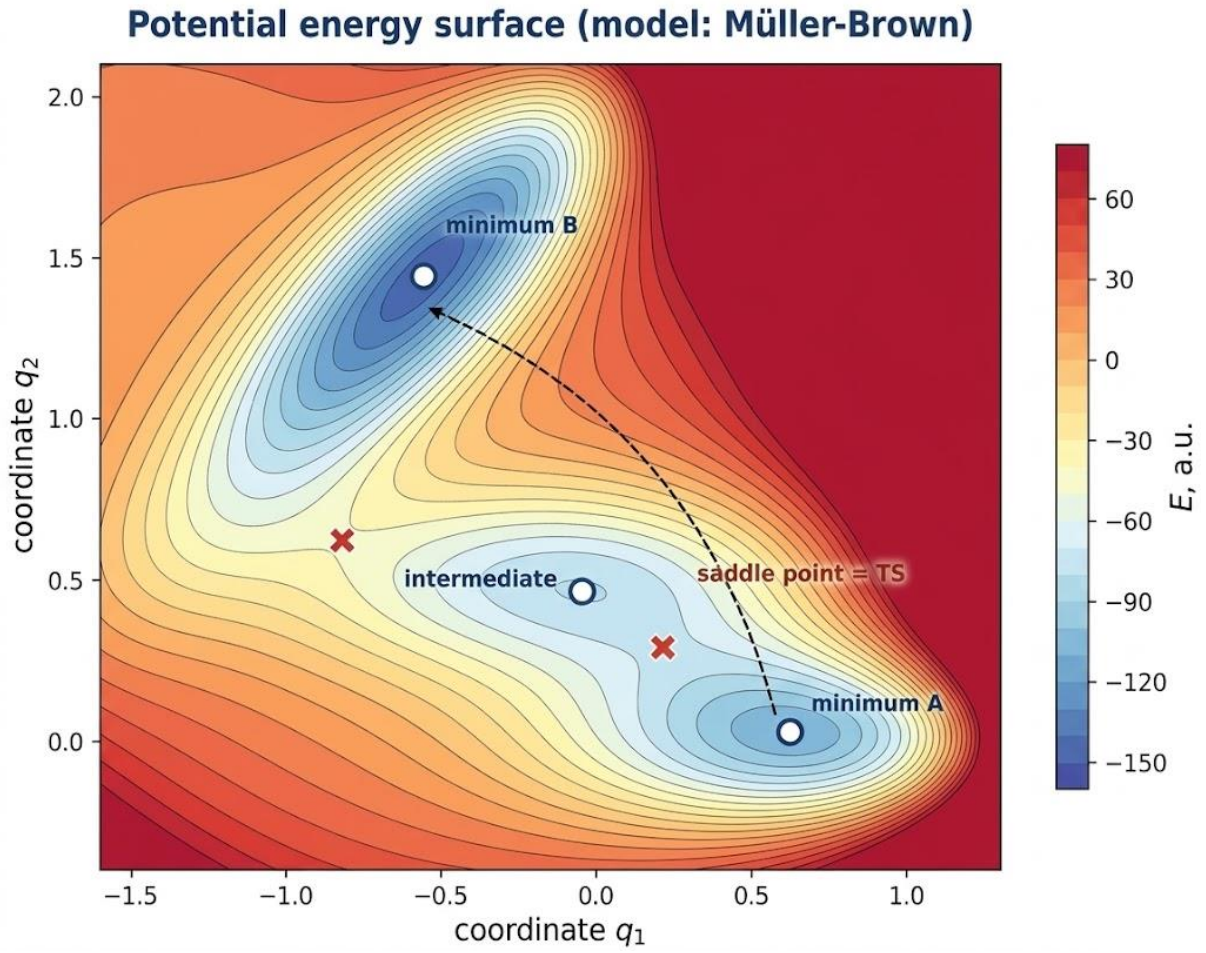


Fig. 27. A model potential energy surface.

The young man: So, the reaction is a path through the valley from one minimum to another over the pass.

Sage: The best path is called **the minimum energy path (IRC).** How can one distinguish a minimum from a saddle point in calculations? The second derivatives of energy are calculated **— the vibrational** frequencies. At a minimum, all frequencies are real. At a saddle point, one frequency **is imaginary:** it corresponds to the motion along the reaction path through the saddle.

> **[UV] Transition state criterion**
> The transitional state is a stationary point with exactly one imaginary frequency. The presence of two or more imaginary frequencies indicates that a pass has not been found, but rather a peak or a more complex point.

Example: H + H2 → H2 + H

Sage: The simplest reaction. The calculation yields a linear transition state H···H···H with a barrier of about 40 kJ/mol (0.42 eV). The imaginary frequency is about 1500 cm−1 (in the same units — 1500i). From experiments, the activation energy is lower — around 30 kJ/mol, because zero-point vibrations and proton tunneling are taken into account.

Young man: I will evaluate the Wigner tunneling correction: $\kappa = 1 + (1/24)\cdot(h\nu^{\ddagger}/kBT)^2$. kT at 298 K = 207 cm^{-1}. $h\nu^{\ddagger}/kT = 1500/207 \approx 7.2$. $\kappa = 1 + 7.2^2/24 = 1 + 2.2 = 3.2$. The rate is three times higher!

> **[OD] Wigner correction — rough**
> For high barriers and large $h\nu^{\ddagger}/kT$, the Wigner correction is either underestimated or overestimated; exact calculations use, for example, tunneling through small curvature. It is suitable for order-of-magnitude estimates.

From calculation to the rate constant

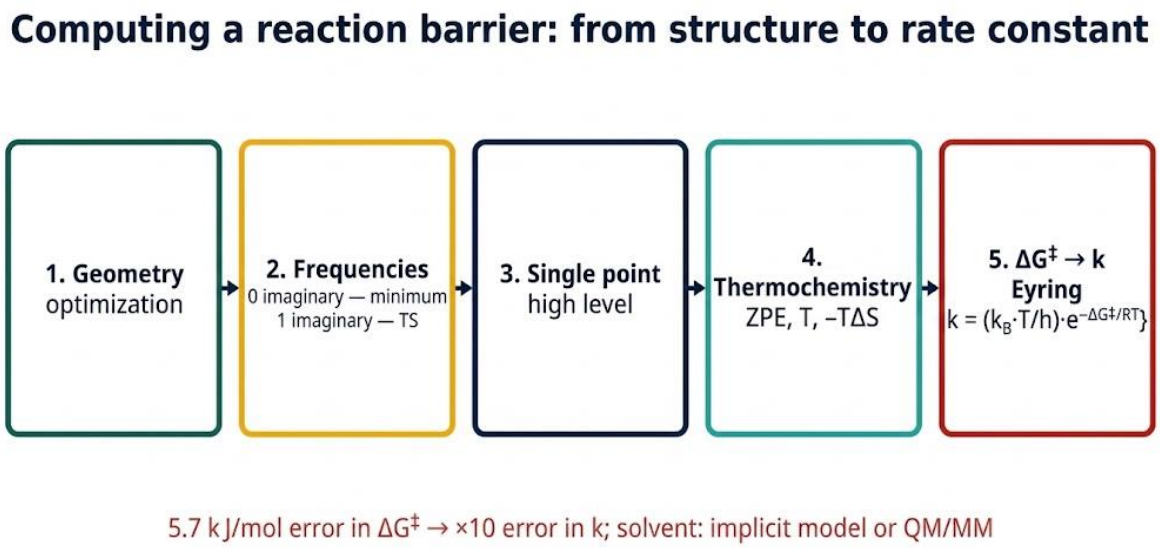


Fig. 28. The workflow for computing a reaction barrier.

Sage: The workflow is as follows: find the structures of the reactant and the transition state; verify by frequencies that this is a minimum and a saddle point; calculate the energy at a high level; add thermochemical corrections (zero-point vibrations, temperature corrections, entropy); obtain ΔG‡ and substitute it into the Arrhenius equation.

Young man: And what accuracy is needed? From table 15: 5.7 kJ/mol — this is exactly a tenfold error in k. And the DFT errors are those same 5–15 kJ/mol.

Table 19. The price of a barrier error at 298 K.

Error in ΔG‡, kJ/mol	Error in k (factor)
2	2,2
4	5
8	25
12	≈ 130

Sage: Therefore, calculations are more reliable **for** comparisons (which catalyst is better, which pathway is faster — errors are partially reduced) than for absolute rates. The modern approach is to combine fast DFT for searching and accurate CCSD(T) for verifying key points, and in recent years — to use potentials trained on machine learning calculations.

> **[OD] Limits of Possibilities**
> The calculation on "pure" models (molecule in vacuum, ideal crystal) does not take into account the solvent, defects, and the dynamics of the real surface. The solvent is considered using implicit models, while enzymes and large systems are treated with a hybrid of quantum and classical mechanics (QM/MM); Karplus, Levitt, and Warshel received the Nobel Prize in 2013 for this.

Calculation and catalysis

Young man: Is it possible to calculate the "volcano" Sabathier from chapter 9?

The sage: Yes, and this is one of the main successes. Nørskov and colleagues showed that the adsorption energy of hydrogen on the surface ΔGH serves as a good descriptor for hydrogen production: the best catalyst has ΔGH ≈ 0. The calculation gives about −0.09 eV for platinum, about −0.27 eV for nickel, and positive values for gold and silver. The calculations also predict new materials, which are then tested in the laboratory.

> **[OD] How to test hypotheses through calculation**
> If the acceleration of the process is explained by tunneling (including in the author's hypothesis of ITÉ, chapter 8), the calculation provides verifiable numbers: the height and width of the barrier at a specific section of the surface, the imaginary frequency, the tunneling coefficient, and the predicted isotopic effect H/D. The agreement or discrepancy with the experiment determines whether an additional mechanism is needed. The conclusion about the existence of such a mechanism remains a hypothesis [G] until this verification.

Chapter summary

- A reaction is a path on the PES between minima through a saddle point with one imaginary frequency.
- ΔG‡ from [a] calculation → k by Eyring; a 5.7 kJ/mol error gives ×10.
- Calculations are stronger for comparison and understanding of the mechanism than for absolute rates.

Self-check

- What conclusion can be drawn about the stationary point if the frequency calculation yields two imaginary frequencies?
- What is the Wigner tunnelling factor for ν‡ = 1000i cm−1 at 298 K? (≈ 2)

Glossary

Russian	Английский
potential energy surface	потенциальная энергия поверхности
saddle point	седловая точка
imaginary frequency	мнимая частота
intrinsic reaction coordinate (IRC)	реакционный путь (IRC)
tunneling coefficient	tunnelling coefficient
QM/MM	QM/MM

Chapter 12. Spectra and Photochemistry

In this chapter

Vibrational spectra and force constants · Beer-Lambert-Bouguer law · Franck-Condon principle · Jablonski diagram · photon energy and binding energy · quantum yield (including > 1) · photocatalysis on semiconductors, band edges, and efficiency.

Vibrational spectra and force constants · the Beer–Lambert law · the Franck–Condon principle · the Jablonski diagram · photon energy and bond energy · quantum yield (including > 1) · semiconductor photocatalysis, band edges and efficiency.

Young man: Sage, spectra are the "fingerprints" of molecules. But what exactly do we learn from them?

Sage: The forces of bonds and the positions of levels. A simple example is the oscillation of a bond as a spring: ν□ = (1/2πc)·√(k/μ), where k is the force constant, μ is the reduced mass. For HCl k = 516 N/m, μ = 0.980 a.u.m. Calculate.

Young man: μ = $^{0.980\cdot1.66\cdot10-27}$ = 1.627·10−27 $^{kg.}$ √(k/μ) = √(516/1.627·10−27) = 5.63·10^14 s−1. I divide ^{by} 2πc = 1.884·10^11 cm/s: ν□ ≈ 2990 cm−1. This is the infrared band HCl!

Sage: And for deuterated DCl, the adjusted mass is 1.906. How will it change?

Young man: ν□ ∝ μ−1/2: 2990·√(0.980/1.906) = 2990·0.717 = 2144 cm−1. Exactly that is observed. This is the same effect that gives the isotopic effect of speed in chapter 8: the zero energy is lower for the heavy isotope.

[UV] Oscillations and Thermodynamics

The frequencies of oscillations from the spectra determine the vibrational heat capacity (chapter 1: "frozen" degree of freedom with θкол = hcν□/k ≈ 6300 K for H2), entropy, and zero-point energies — that is, thermodynamic functions and equilibrium constants can be calculated from the spectra. This is the foundation of statistical thermodynamics.

Light absorption: Beer's law

Sage: For electronic transitions, the intensity of absorption is described by the Beer-Lambert law: A = ε·c·l, where ε is the molar absorptivity, l is the thickness of the layer. For KMnO4 at 525 nm, ε ≈ $^{2400\ M-1\cdot cm-1.}$

Young man: A solution ^{of} 10−4 M in a 1 cm $^{cuvette:}$ A = 2400·10−4·1 = 0.24. 1 − $^{10-0.24}$ = 43% of light is absorbed. This law is the basis of photometry and kinetics by spectra (chapter 7).

What happens after absorption

Sage: The electronic transition occurs in ≈ 10−15 ^{s} — the nuclei do not have time to shift (Franck–Condon principle): the transition is "vertical," as shown in the figure. Then the molecule returns through various pathways.

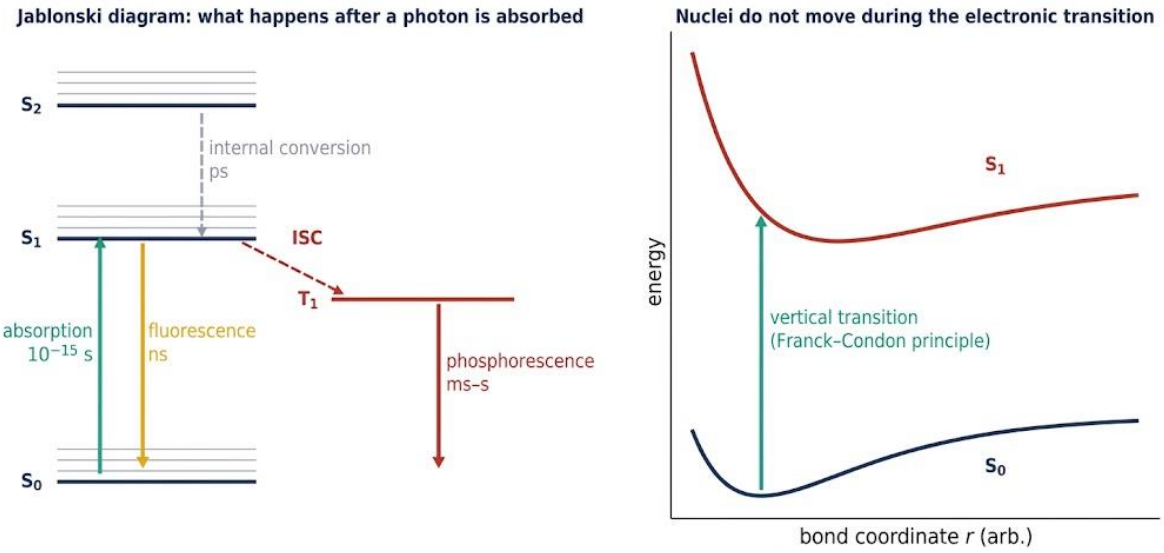


Fig. 29. The Jablonski diagram and the Franck–Condon principle.

Table 20. Time scales after photon absorption.

Process	Time	Comment
absorption	~10–15 s / s	nuclei frozen
internal conversion	ps / ps	heat, no light
fluorescence	ns / ns	$S_1 \rightarrow S_0$ with emission
intercombination conversion (ISC)	ps–ns / ps–ns	$S_1 \rightarrow T_1$, spin flip
phosphorescence	ms–s / ms–s	$T_1 \rightarrow S_0$, spin-forbidden

Sage: Now the main photochemical law: one absorbed quantum excites one molecule (Stark–Einstein). The energy of a mole of quanta $E = N_A hc/\lambda \approx 119{,}600$ kJ·nm/λ.

Table 21. Photon energy.

λ, nm / nm	Region	E, eV / eV	E, kJ/mol
250	UV / UV	4,96	479
400	violet	3,10	299
500	green	2,48	239
700	red	1,77	171
1000	near IR	1,24	120

Young man: The C–C bond is 347 kJ/mol, O–H is 463. Thus, ultraviolet light at 250 nm (479) can break almost any bond, while red light (171) cannot. And the ozone layer? O_2 breaks at 498 kJ/mol: λ = 119 600/498 = 240 nm. It is this UV that transforms O2 into oxygen atoms, from which ozone is formed!

Quantum output: can it be greater than one?

Sage: The quantum yield Φ = the number of molecules that have reacted (or formed) per absorbed quantum. For a simple reaction, $\Phi \leq 1$, but for a chain reaction, it can be enormous. For H2 + Cl2 → 2HCl in light, Φ reaches 10^5–10^6: one quantum generates a Cl· atom, which initiates a chain of hundreds of thousands of links (chapter 9).

[OD] Definitions of quantum yield
The "apparent quantum yield" 2N(H2)/N(photons) is often used for the complete decomposition of water, because two electrons are needed per molecule of H2. With this formula, the maximum is equal to 1, while when counting "molecules per quantum" it is 0.5. Any number Φ makes sense only when specifying the formula and the method of measuring the absorbed light (chapter 6, table 11).

Photocatalysis on semiconductors

The sage: In a semiconductor, a photon with E > Eg transfers an electron from the valence band to the conduction band, leaving a "hole." Two conditions are needed for water decomposition: the bottom of the conduction band must lie above (more negative than) the potential H+/H2 (0 V), and the top of the valence band must be below (more positive than) the potential O2/H2O (+1.23 V). Thus, Eg must be greater than 1.23 eV, in practice around 1.8–2.0 eV considering overpotentials.

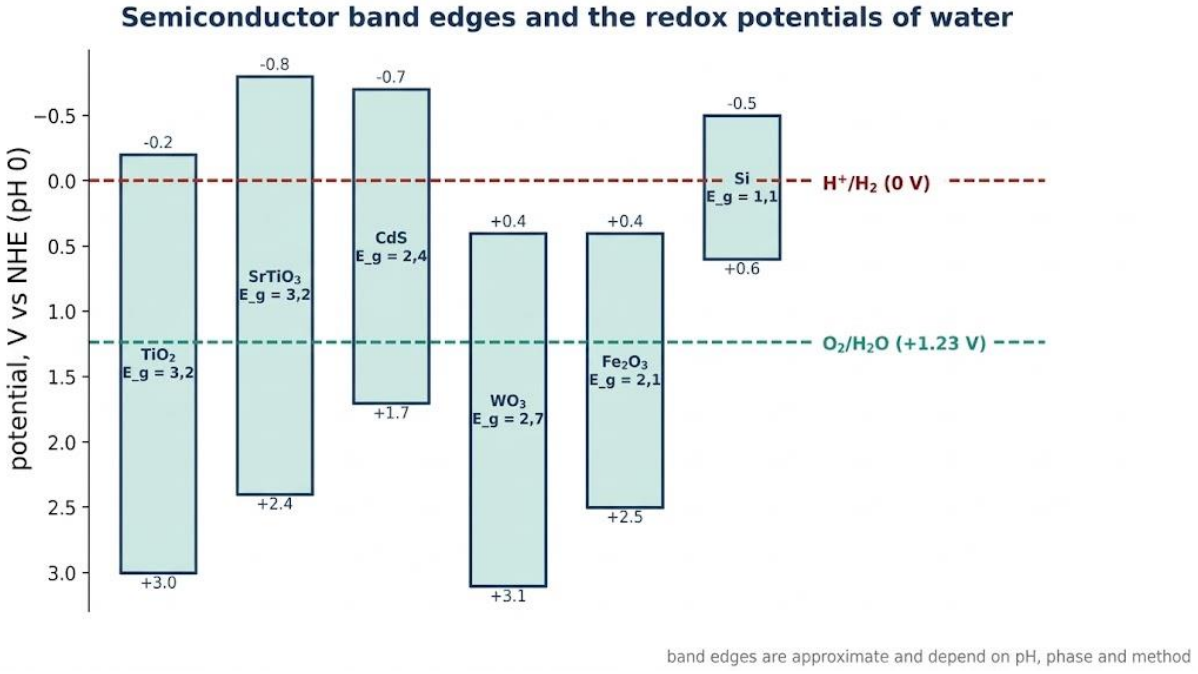


Fig. 30. Semiconductor band edges and water potentials.

Young man: TiO2: Eg = 3.2 eV, which means it only works in light shorter than 1240/3.2 = 387 nm — this is ultraviolet, only about 5% of the energy of sunlight. And Fe2O3 absorbs visible light (2.1 eV), but its conduction band lies below 0 V: it cannot restore protons.

Sage: Correct. Efficiency is assessed through STH — the fraction of solar energy stored in hydrogen: STH = j·1.23 V/Pсвет, where j is the photocurrent. What current is needed for 10% under standard sunlight of 100 mW/cm²?

The current density j = 0.10·100 mW/cm²/1.23 V = 8.1 mA/cm². Then the rate H2 = j/(2F) = 8.1·10⁻³/(2·96 485) = 4.2·10⁻⁸ mol/(cm²·s), about 3 g of hydrogen per m² per hour.

[UV] DFT and band gap width
Standard DFT (GGA) approximations underestimate the band gap of semiconductors, often by a third or more. Hybrid functionals and GW-type methods correct this, but are more expensive. Therefore, it is important to know which functional has been used for the calculation of photocatalysts.

[NI] Data on steam photocatalysis and "energy 12.6 eV"
In Chapter 6, we discussed the data reported by the author (Φ ≈ 1 at an effective energy of 12.6 eV, completeness 85–95%, efficiency 13.5–20% at 350 and 450 °C). Here we add that 12.6 eV corresponds to a wavelength of 1240/12.6 = 98 nm — vacuum ultraviolet, and is close to the first ionization energy of water (12.62 eV). The definition of "effective energy" of an event does not have to coincide with the energy of the photon; this definition needs to be explicitly stated. The data remain the result of one group; control experiments (Chapter 6, Table 11) will address the issue of independent confirmation.

Chapter summary

- Spectra give force constants and levels; $\nu\square \propto \sqrt{(k/\mu)}$.
- After absorption: Franck–Condon, internal conversion, fluorescence, intersystem crossing, phosphorescence.
- Φ can exceed 1 (chains); in photocatalysis, the band edges, Eg, and the definitions of Φ and η are determined.

Self-check

- Compute $\nu\square$ for k = 500 N/m, μ = 1 amu. (≈ 2900 cm−1)
- What wavelength can break a C–C bond (347 kJ/mol)? (≈ 345 nm)

Glossary

Russian	Английский
the Beer-Lambert law	Beer–Lambert law
Franck–Condon principle	Franck–Condon principle
fluorescence	fluorescence
phosphorescence	фосфоресценция
quantum output	квантовый выход
restricted area	запретная зона
photocatalyst	фотокатализатор

Final Conclusion

We started with four questions: whether it **will go,** how **far, how fast, and why. It is now** clear that these are four sides of the same picture: from the properties of a single molecule to the behavior of a mole.

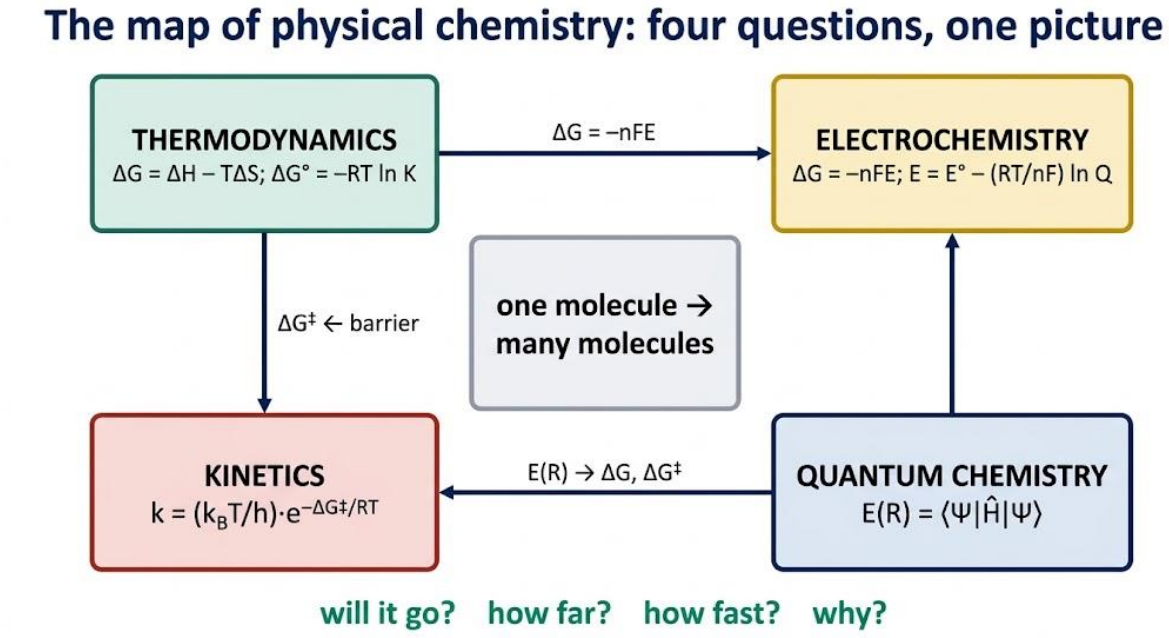


Fig. 31. The map of physical chemistry.

Key formulas

Table 22. Key formulas of the book.

Idea	Formula	Chapter / Ch.
first law	$\Delta U = Q + w$	1
entropy	$dS = \delta Q_{обр}/T;\ S = kB \cdot \ln W$	2
free energy	$\Delta G = \Delta H - T\Delta S;\ \Delta G^\circ = -RT \cdot \ln K$	3
dependence of K on T / K versus T	$d \ln K/dT = \Delta H^\circ/(RT2)$	3
phase rule	$F = C - P + 2$	4
osmosis	$\Pi = c \cdot R \cdot T$	4
activity	$\lg \gamma_{\pm} = -0.509 \cdot \lvert z_+ z_- \rvert \cdot \sqrt{I}$	5
EMF / ЭДС	$\Delta G = -nFE;\ E = E^\circ - (RT/nF) \cdot \ln Q$	6
overpotential	$\eta = a + b \cdot \lg j;\ b = 2{,}303RT/(\alpha F)$	6
speed / rate	$v = k[A]m[B]n;\ t_{1/2} = \ln 2/k$ (1st)	7
Arrhenius	$k = A \cdot \exp(-Ea/RT)$	8
Eyring	$k = (kBT/h) \cdot \exp(-\Delta G^\ddagger/RT)$	8
Michaelis–Menten	$v = Vmax[S]/(Km + [S])$	9
Born–Oppenheimer	E(R) — potential energy surface / PES	10–11
Beer–Lambert	$A = \varepsilon \cdot c \cdot l$; Φ = события/кванты / events/photons	12

Ten corrections

Table 23. The book's ten corrections.

№	Common claim	Chapter / Ch.
1	Heat is contained in a body	1
2	"Entropy is just disorder"	2
3	"Life violates the second law"	2
4	"ΔG < 0 means fast"	3, 8
5	"Skates glide because ice melts under the blade"	4
6	"Ions in solution act independently"	5
7	"Reaction order equals the sum of coefficients"	7
8	A catalyst shifts the equilibrium	9
9	"Efficiencies of any methods can be compared without definitions"	6, 12
10	"Ea is the height of the threshold that the molecule simply jumps over"	8, 11

What remains open

- Why the Universe began with low entropy is a hypothesis [Г]; the mechanism of gliding on ice is not fully established.
- The accuracy of DFT functionals and absolute rates remain a challenge: a 5.7 kJ/mol error is ×10.
- The ITE hypothesis and the steam-photocatalysis data are [Г]/[НИ]: controls, isotope labelling, a full energy balance and independent replication are needed.

To the reader

Physical chemistry is the ability to ask questions with numbers: "How much? At what temperature? With what error? What does this follow from?" If you have learned to verify a formula by calculation, compare the calculation with a reference book, and honestly record definitions, you have mastered the main aspect of this science. The next steps are statistical thermodynamics, nonequilibrium processes, spectroscopy, and computational chemistry: the same questions, but with even deeper answers.

Let the number in your hand always be accompanied by the question: what does it mean and how can it be verified?

Afterword

This book is conceived as a logical continuation of "General and Inorganic Chemistry": there we learned to write ΔH, ΔG, K, E°, and activation energy; here we clarify where they come from, what they mean, and where they cease to be applicable. Below are a few words on how to work with it.

How to read

Table 24. Reading routes.

Route	Chapters	For whom
Short	1, 2, 3, 6, 8, 9	key ideas: energy, entropy, ΔG, electrochemistry, kinetics, catalysis
Full course	1–12 in order	systematic study
With calculations	1, 3, 5, 6, 8, 10, 11	numbers, units, order-of-magnitude estimates
Hydrogen and photocatalysis	3, 6, 9, 12	electrolysis, fuel cells, catalysis, photochemistry

What to check

Numerical data — constants, energies, potentials, model parameters — are given with rounding and correspond to typical reference values; the calculations of the Youth have been recalculated based on this data. For work where accuracy is important, verify the numbers with the current reference guide.

Data on steam photocatalysis and "effective energy of 12.6 eV" (Chapter 6, Table 11; Chapter 12) are marked as [NI]: these are results from one group, and we have not seen independent confirmation. The ITE hypothesis (Chapters 8 and 11) is marked as [G]. These sections are a working example of how to verify new results: through definitions, energy balances, control experiments, and independent reproduction.

Data dependent on the date (benchmarks for the efficiency of electrolysis and photocatalysis, accuracy of calculation methods) is dated 2026 and will become outdated over time.

To the reader

The series is based on the image of a "slacker" — a person who is not afraid to ask "why is it so?" and to verify what has been said. If you found a mistake, inaccuracy, or controversial point in the book — it is the reader's merit: this is how science progresses, through corrections. Report your findings, and we will correct them.

Let the next page of your physical chemistry begin with your own calculation.

Appendix. Consolidated Glossary

The terms are collected from all chapters in alphabetical order of their Russian names; in the last column is the chapter where the term is introduced or discussed in more detail.

Термины из всех глав в алфавитном порядке русских названий; последний столбец указывает главу, в которой термин введен или обсуждается наиболее подробно.

Table 25. Consolidated glossary.

Russian	Английский	Ch.
QM/MM	QM/MM	11
azeotrope	azeotrope	4
activated complex	активированный комплекс	8
activity, activity coefficient	активность, коэффициент активности	5
Born–Oppenheimer approximation	Born–Oppenheimer approximation	10
Beer-Lambert law	Beer–Lambert law	12
Van 't Hoff equation	уравнение ван't Гоффа	3
variational principle	вариационный принцип	10
internal conversion	внутренняя конверсия	12
internal energy	внутренняя энергия	1
the second law of thermodynamics	второй закон термодинамики	2
Gibbs energy	Энергия Гиббса	3
hybrid functional	гибридный функционал	10
Debye length	Дебаевская длина	5
Debye–Hückel theory	Дебай–Гюккель теория	5
diffusion limit	диффузионный предел	7
Henry's law	Закон Генри	4
Kohlrausch's law	Kohlrausch’s law	5
Raoult's law	Рауля закон	4
rate law	скорость реакции	7
restricted area	запретная зона	12
isotopic effect	изотопный эффект	8
integral rate law	интегрированный закон скорости	7
ionosphere	ионная атмосфера	5
ionic strength	ионная сила	5
calorimeter	calorimeter	1
catalyst	катализатор	9
quantum output	квантовый выход	12
colligative properties	colligative properties	4
equilibrium constant	константа равновесия	3
speed constant	константа скорости	7
correlations energy	корреляционная энергия	10

Russian	Английский	Ch.
critical point	критическая точка	4
Lainuiver-Burke graph	Lineweaver–Burk plot	9
Lindemann mechanism	Механизм Линдеманна	9
initial velocity method	метод начальных скоростей	7
Michaelis-Menten equation	Уравнение Михаэлиса–Ментена	9
imaginary frequency	мнимая частота	11
molecularity	молекулярность	9
molar conductivity	молярная проводимость	5
Nernst equation	Уравнение Нернста	6
zero-point energy (ZPE)	ноль-точечная энергия (НТЭ)	10
reciprocal number	число оборота	9
reversible process	обратимый процесс	1
osmotic pressure	осмотическое давление	4
residual entropy	остаточная энтропия	2
overvoltage	overpotential	6
transitional state	переходное состояние	8
half-life	период полураспада	7
density functional theory (DFT)	теория функционала плотности (DFT)	10
potential energy surface	потенциальная энергия поверхности	11
reaction order	порядок реакции	7
phase rule	фазовое правило	4
pre-exponential factor	pre-exponential factor	8
pseudorder	псевдо-порядок	7
reaction coefficient	реакционный коэффициент	3
Sabathier principle	Sabatier principle	9
saddle point	седловая точка	11
reaction rate	скорость реакции	7
stationary approximation	steady-state approximation	9
Tafel equation	Уравнение Тафеля	6
heat capacity	теплоемкость	1
thermoneutral voltage	термонеутральное напряжение	6
exchange current	плотность обменного тока	6
fuel cell	топливный элемент	6
the third law of thermodynamics	third law of thermodynamics	2
triple point	тройная точка	4
tunneling	tunnelling	8
fluorescence	fluorescence	12
phosphorescence	фосфоресценция	12
photocatalyst	фотокатализатор	12
Franck–Condon principle	Franck–Condon principle	12

Russian	Английский	Ch.
state function	функция состояния	1
Hartree–Fock method	Hartree–Fock метод	10
chemical potential	химический потенциал	3
Hückel method	Hückel method	10
Carnot cycle	Карно́тов цикл	2
transfer number	транспортное число	5
EMF	электромоторная сила (ЭДС)	6
Eyring equation	Уравнение Эйринга	8
electrode potential	электродный потенциал	6
electrical conductivity	проводимость	5
activation energy	активационная энергия	8
delocalization energy	делокализационная энергия	10
enthalpy	энтальпия	1
entropy	энтропия	2
efficiency of "sun → hydrogen" (STH)	солнечно-водородная (СВ) эффективность	12

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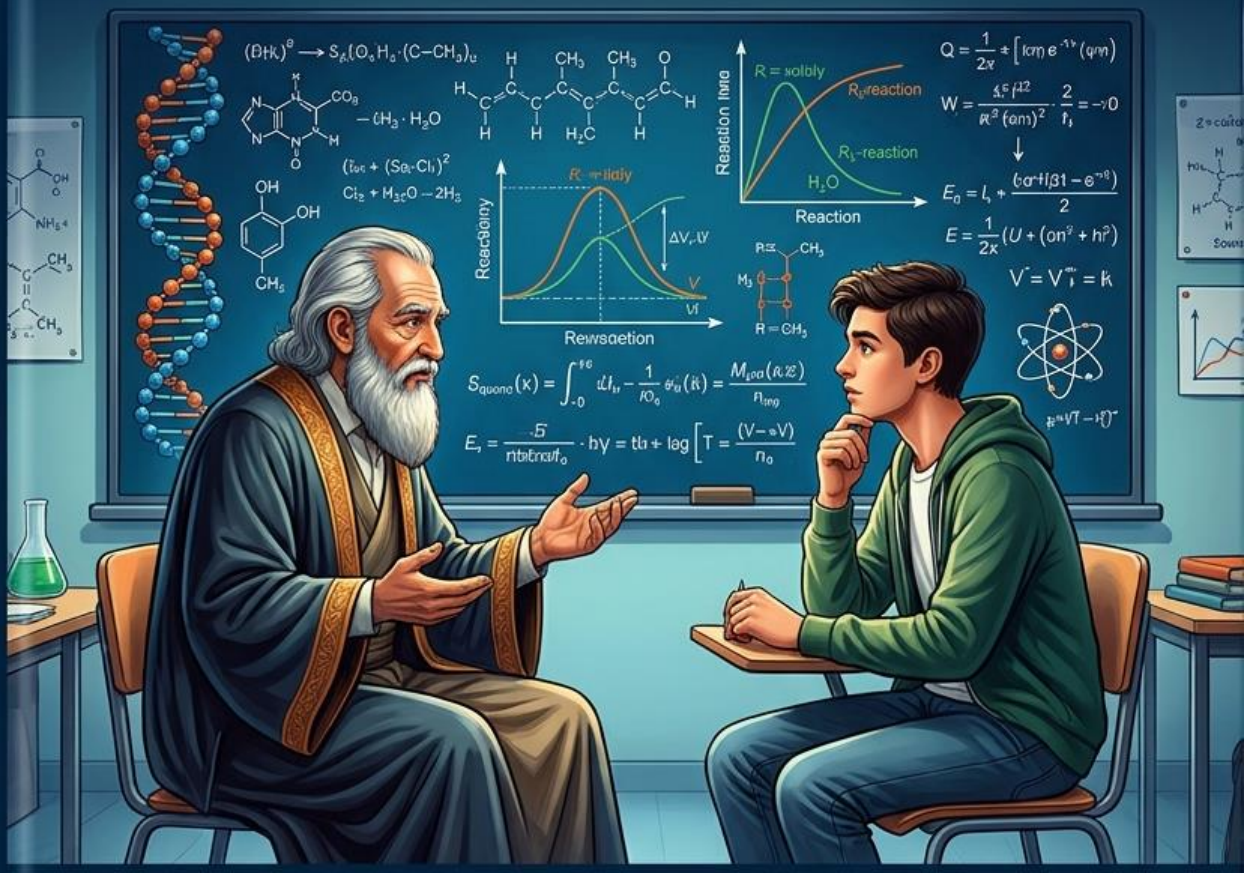
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БЕСЕДЫ МУДРЕЦА И ЮНОШИ

ФИЗИЧЕСКАЯ ХИМИЯ

Physical Chemistry

Рахимов Р.Х.



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12 chapters · 31 figures · 25 tables — in Russian and English

Серия посвящена памяти академика Садыка Азимовича Азимова

Tashkent · 2026